

AP190 PRACTICUM PORTFOLIO

Submitted by:

MIEL GABRIEL LAGDAN

Bachelor of Science in Applied Physics

June 22, 2026 - July 31, 2026

Submitted to:

ERIC M. INOCENCIO, MSc., LPT

OJT Coordinator

Physics and Geology Unit

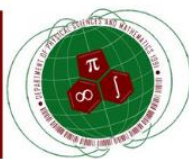
Department of Physical Sciences and Mathematics

University of the Philippines Manila

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Course Description



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AP 190 PRACTICUM GUIDELINES FOR STUDENTS AND PROSPECTIVE SUPERVISOR

A. INTRODUCTION

What is the BS Applied Physics (Health Physics) program about?

The BS Applied Physics (Health Physics) program at the University of the Philippines Manila is a four-year undergraduate degree program that combines a strong foundation in physics with a focus on applications in the field of health and medical physics. The program provides students with a broad range of theoretical knowledge and practical skills related to radiation safety, medical imaging, radiation therapy, and nuclear medicine.

In particular, the program emphasizes the use of physics principles and techniques to understand the behavior of radiation in biological systems and the application of this understanding to the diagnosis and treatment of diseases. The program also emphasizes the importance of safety and regulatory compliance in the use of radiation in medical applications.

The program consists of a mix of classroom instruction, laboratory work, and hands-on experience through internships and research projects. Graduates of the program are prepared for careers in a range of fields, including medical physics, health physics, radiation safety, nuclear medicine, and radiation therapy, as well as for further study in graduate programs in physics, medical physics, or related fields.

What is the significance of the internship program?

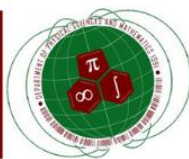
The internship program is a crucial component of the BS Applied Physics (Health Physics) program at the University of the Philippines Manila. It provides students with an opportunity to apply their theoretical knowledge and skills in real-world settings, gain practical experience, and develop a deeper understanding of the field of health and medical physics.

Through internships, students can gain exposure to a range of professional settings, including hospitals, research laboratories, and regulatory agencies, and work alongside professionals in the field. This exposure provides them with valuable networking opportunities and can help them to identify potential career paths.

The internship program also enables students to develop important professional skills, such as communication, collaboration, and critical thinking, which are essential for success in the field of health and medical physics. Moreover, the internship program helps students to identify areas of interest within the field and to refine their career goals.

Overall, the internship program is an important aspect of the BS Applied Physics (Health Physics) program, as it helps to prepare students for successful careers in the field of health and medical physics and contributes to the development of a highly skilled and knowledgeable workforce in this critical area of healthcare.





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B. OBJECTIVES

At the end of the program, the student should be able to deliver the following:

BS APPLIED PHYSICS (HEALTH PHYSICS) PROGRAM LEARNING OUTCOMES:

- Apply theoretical, computational, and experimental physics in a workplace;
- Conduct research related to health and medical physics that contribute to national development;
- Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
- Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
- Synthesize and effectively communicate scientific information in both English and Filipino; and
- Act in recognition of professional, social, and ethical responsibility.

with the following **AP 190 PRACTICUM COURSE LEARNING OUTCOMES (CLO):**

1. Demonstrate an understanding of theoretical knowledge on health physics by performing their duties as a student intern on their chosen institution for at least 200 hours
2. Exemplify professionalism and work ethical standards with different radiation personnel whether in a hospital or an industrial setting
3. Appreciate health physics as a developing field in the country
4. Create a portfolio summarizing the experience of a health physics intern in the field (hospital, research, or industrial setting)

C. General Guidelines

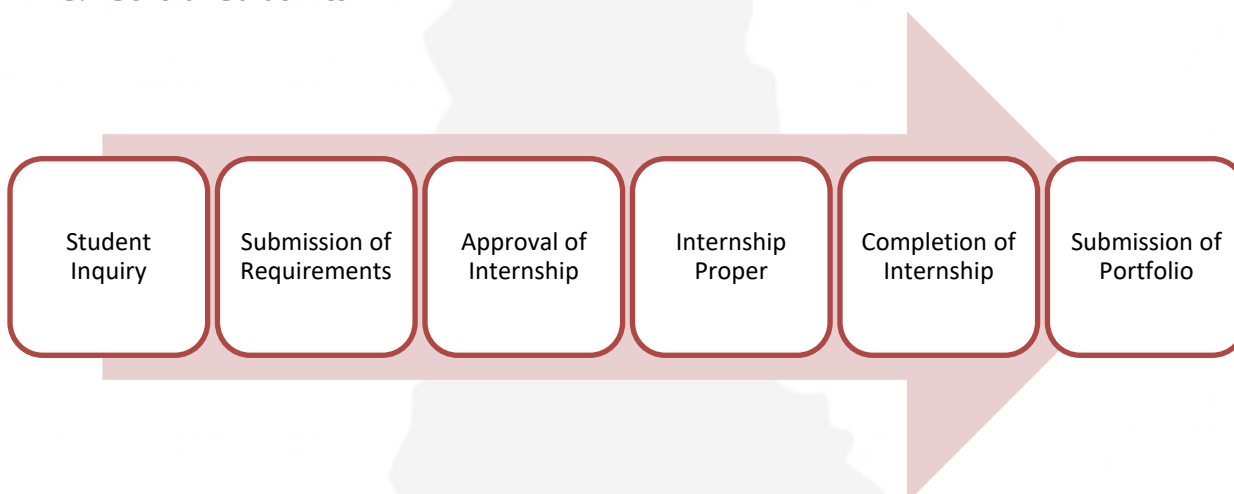
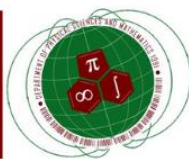


Figure 1: Flowchart of the entire process from application to completion of practicum/internship.

1. Students will research which institutions accept internships and choose one among them to apply for an internship position.
2. Students will collect the required documents by the partner institution (the accepting institution) and submit.





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- The supervisor will submit a fully accomplished and signed acceptance letter form, indicating the approval of the student internship, with some ideal details of what the training program will be. **(Letter of Acceptance Form and Attachments)**
- The student intern will fulfill their duties accordingly.
- Upon completion of the 200-hour training, the supervisor should provide the student with their Certificate of Completion, signed by the supervisor and/or other authorities in the partner institution. Certificate Template is provided by the institution or the OJT-Coordinator if requested.
- Students will submit their portfolio to the OJT-Coordinator, with a copy of the Completion Certificate attached in the Appendices.

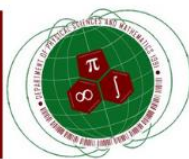
D. Grading System:		Supervisor will give 70% of the grade based on the student's performance using the evaluation sheet to be provided at the last week of the approved training schedule. <i>(attached from the letter of approval form)</i> OJT Coordinator will give 30% of the grade based on the student's submissions (portfolio and daily journal) and one-time Supervisor-OJT coordinator meeting (via <u>online or in-person</u>) to be scheduled after the first half of the approved training schedule
Supervisor:	70%	
OJT Coordinator:	<u>30%</u>	
Total of	100%	
Passing total percentage: 75%		

E. List of Expectations/ Possible Task to be delivered by the student

For the guidance of the supervisors, below are the list of expectations/possible tasks to be delivered by the students. These are just recommendations and to give the supervisor an overview of what specific duties and responsibilities should the student accomplish throughout the training. Still, the **supervisor can deviate from the list** whenever needed. The supervisor and student **can make use** the table below as a **checklist**.

CLO1	Demonstrate an understanding of theoretical knowledge on health physics by performing their duties as a student intern on their chosen institution for at least 200 hours
	Assist in the calibration and maintenance of radiation detection equipment, such as Geiger counters, scintillation detectors, and ionization chambers.
	Participate in the inspection and evaluation of radiation safety programs and practices in various settings, such as hospitals, research laboratories, and industrial facilities.
	Assist in the planning and execution of radiation safety training programs for staff and other personnel.
	Help in the collection and analysis of radiation exposure data, including environmental monitoring data, worker exposure data, and patient exposure data.
	Participate in the design and implementation of radiation protection programs, such as radiation shielding, contamination control, and waste management.

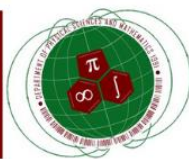




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	Assist in the development and maintenance of radiation safety policies, procedures, and regulations, in compliance with national and international standards.
	Help in the preparation and review of safety reports, safety plans, and safety audits related to radiation safety.
	Participate in the investigation of radiation incidents and accidents, including root cause analysis, corrective actions, and follow-up activities.
	Assist in the evaluation and selection of radiation protection equipment and materials, such as shielding materials, personal protective equipment, and dosimetry devices.
	Participate in the research and development of new technologies and techniques in health and medical physics, such as imaging techniques, radiation therapy, and nuclear medicine.
CLO2	Exemplify professionalism and work ethical standards with different radiation personnel whether in a hospital or an industrial setting
	Attend safety meetings and training sessions regularly to stay informed and updated on safety policies, regulations, and best practices.
	Observe and follow safety rules and procedures in the workplace, such as wearing personal protective equipment, maintaining good hygiene, and following proper waste management practices.
	Practice good communication skills when working with radiation personnel and other staff, such as listening actively, asking questions, and providing feedback.
	Maintain a positive attitude and a professional demeanor at all times, even in challenging situations, and work collaboratively with others to achieve common goals.
	Respect the confidentiality of sensitive information, such as patient records and proprietary data, and follow data privacy policies and regulations.
	Adhere to ethical standards of conduct and behavior, such as honesty, integrity, and respect for diversity, and report any violations or concerns to the appropriate authorities.
	Demonstrate a commitment to lifelong learning and professional development by seeking out opportunities to acquire new knowledge and skills, attending conferences and workshops, and networking with other professionals in the field.
	Participate in community service activities or outreach programs related to radiation safety and health physics, such as public lectures, seminars, or health fairs.
	Take responsibility for one's own safety and well-being, as well as that of others in the workplace, by reporting any unsafe conditions or incidents promptly and following appropriate protocols for reporting and investigating incidents.
	Maintain accurate and detailed records of work activities, such as experimental data, safety inspections, and incident reports, and communicate results and findings effectively to colleagues, supervisors, and other stakeholders.
CLO3	Appreciate health physics as a developing field in the country
	Attend seminars, conferences, and other events related to health physics to learn about the latest developments and trends in the field.
	Conduct research on the current state of health physics in the Philippines, including the challenges and opportunities for growth and development.
	Explore the different applications of health physics in various industries and sectors, such as healthcare, nuclear energy, and radiation safety.





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	Interview professionals in the field, such as health physicists, radiologic technologists, nuclear medicine technologists, and radiation safety officers, to gain insights into their work and experiences.
	Engage in discussions with peers and mentors about the role of health physics in promoting public health and safety, and the importance of promoting awareness and education on radiation safety.
	Visit institutions and facilities that use radiation and nuclear technology, such as hospitals, research laboratories, and power plants, to observe how health physics is applied in practice.
	Participate in outreach programs and advocacy campaigns related to health physics, such as radiation safety awareness campaigns, public lectures, or community health programs.
	Learn about the regulations and guidelines governing the use of radiation and nuclear technology in the Philippines, such as the Food and Drug Administration's (FDA) Center for Device Regulation, Radiation Health and Research (CDRRHR) and the Philippine Nuclear Research Institute (PNRI).
	Develop a deeper understanding of the ethical and social implications of using radiation and nuclear technology in society and engage in discussions on how to balance the benefits and risks of these technologies.
	Reflect on one's own experiences and learning and identify areas for further growth and development in the field of health physics.
CLO4	Create a portfolio summarizing the experience of a health physics intern in the field (hospital, research, or industrial setting)

F. Training Design

- The supervisor has the option to choose from the following training design:
 - Full in-person practicum
 - Hybrid in-person/online practicum (preferably higher percentage of reporting in the field)
- The **full in-person** practicum design is a 100% face-to-face delivery of the training.
- The **hybrid design** allows both the supervisor and the student to have flexible delivery of the training. For example, the student can be in-person reporting from Monday-Wednesday for a certain agreed number of duty hours, and then the rest of the week will be a work-from-home set up, doing tasks that can be done remotely. The online monitoring will be up to the supervisor. This option is also in accordance with the COVID19 protocols of **social distancing**. Preferably, the percentage of in-person training (60%) is higher than the time spent on a work-from-home setup (40%). Again, the supervisor may deviate from this recommendation.

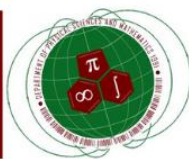
G. Scheduling

By recommendation, the supervisor can choose from the schedule types:

Requirement: 180 hours of training	No of working hours a week	No of working hours a day	No of weeks for completion
Schedule A	40	8	4.5
Schedule B	30	6	6
Schedule C	20	4	9

Note: The **remaining 20 hours** will be spent by the student in **creating their portfolio** and other individual





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submissions to the OJT-coordinator. Only **180 hours** will be given as **contact time** of the student with the supervisor. (**180 hours supervisory tasks + 20 hours of OJT-coordinator tasks = 200 hours**)

The **supervisor may deviate** from the recommendation above. Please note that some students are enrolled in one or two more subjects aside from their practicum. Hence, an additional one buffer week is requested to finish the required hours of training. Moreover, training **design B** (hybrid training) is suggested to make the time flexible for both students and supervisors, preventing unnecessary extension of the internship. However, if there will be unforeseen circumstances, if the student cannot finish the internship on or before the last day of the midyear term (when the students are enrolled for practicum) the **internship extension may be granted** if there will be a **letter of agreement** from the supervisor and OJT-coordinator. This letter will be the replacement for the completion certificate needed in the portfolio, as a partial requirement of the course.

H. Expected **STUDENT** output

Requirement	Submitted to	When
Daily Journal	OJT-coordinator	Weekly
Portfolio (with pictures) in PDF format	OJT-coordinator	At the End of the MidYear Term
Feedback form	OJT-coordinator (required) and/or Supervisor (optional)	At the End of the MidYear Term

I. Requested documents from the **SUPERVISOR**

Requirement	Submitted to	When
Signed Letter of Acceptance Form	OJT-coordinator	Before the internship proper
Signed Certificate of Completion/ Letter of Extension	OJT-coordinator	At the End of the MidYear Term
Accomplished Evaluation Form for the Student	OJT-coordinator (required) and/or Student (optional)	At the End of the Internship Program

Noted by:

Eric M. Inocencio, MSc.

OJT-Coordinator (BS Applied Physics Program)

Department of Physical Sciences and Mathematics, College of Arts and Sciences

University of the Philippines - Manila

Marie Josephine M. De Luna, Ph.D.

Department Chair

Department of Physical Sciences and Mathematics, College of Arts and Sciences

University of the Philippines - Manila



Training Portfolio

2.1 Daily Journals

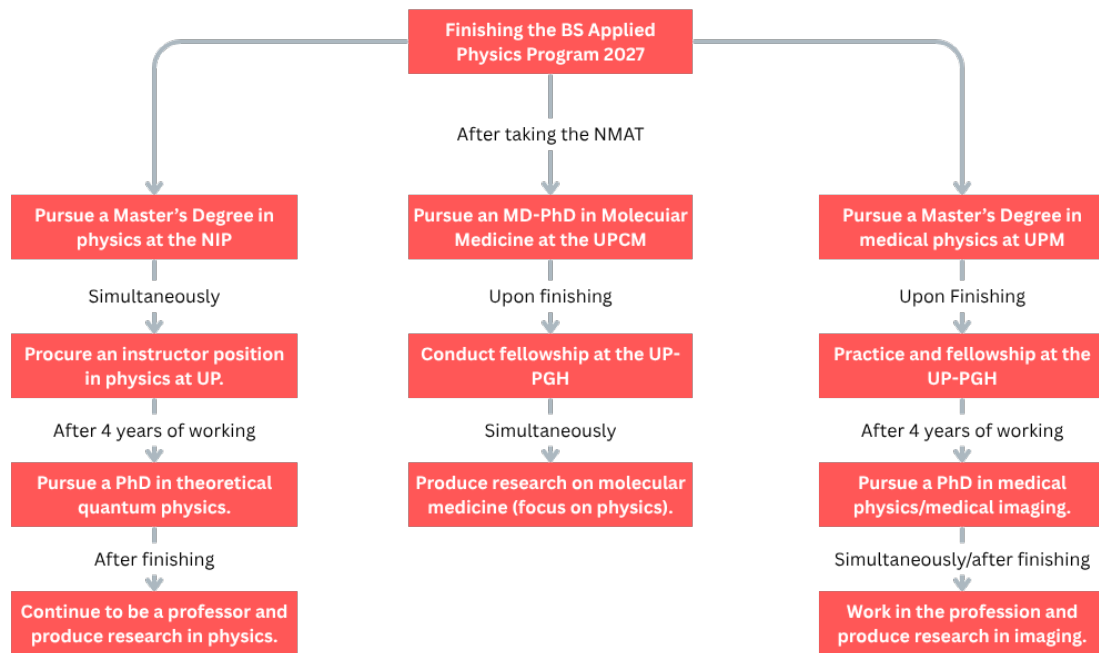
This portion provides the daily journals accumulate during the internship dated June 22, 2026 to [insert end date here]. The following gives an outline of the main progress done.

Date	Progress
June 22, 2026	Orientation
June 23, 2026	B0: Introduction to Paper (Bagunu and Galapon (2025))
June 24, 2026	B0: Primary Reading of Whole Paper.
June 25, 2026	B1: Start of Analysis and Derivations.
June 26, 2026	B1: Research on the Weyl, simplest-Symmetrization, and Born-Jordan ordering rules. Provided short report on progress to the paper to supervisor.
June 29, 2026	B1: Derived relationship of the Schrodinger and Heisenberg pictures of quantum mechanics.
June 30, 2026	B2: Derivation of the images from the Born-Jordan quantization.
July 1, 2026	B3: Continuation of the derivation of the images from the Born-Jordan quantization.
July 2, 2026	B4: Derivation of the images as a result of the modifications to the differential quotient of first type.
July 3, 2026	B5: Suzuki's analysis on Banach spaces.
July 7, 2026	B5: Provision of the rigorous proof for the application of the differential quotient of the second type to operators.
July 9, 2026	B5: Derivation of the theorems (1).
July 13, 2026	B5: Derivation of the theorems (2).
July 14, 2026	B5: Proof for Euler's reflection formula and its relationship to the gamma and beta functions.
July 15, 2026	B5: Derivation of the theorems (3).
July 16, 2026	BF: Initial report/partial report to Dr. Ralph Ferrales.
July 17, 2026	Exploration of the derivation of the Weyl ordering.
July 18, 2026	Reading of Domingo's paper (2016) on Algebra of Non-Weyl operators.
July 19, 2026	B6: Quantization of the Hamilton Equations of Motion
July 20, 2026	BA: Exploration of case for the harmonic oscillator.
July 21, 2026	Further research on the generalized Weyl transform.
July 22, 2026	BF: Completion of the Paper Analysis.
July 23, 2026	BF: Report of the Whole Paper.

Table 1: The bold sections refer to sections worked on the paper. Please refer to Appendix 3.8.

2.2 On-The-Job Training Evaluation

2.3 Personal Professional Plan



Appendix

3.1 Completion Certificate

3.2 Letter of Acceptance

Please see attached file on next page.



NATIONAL INSTITUTE OF PHYSICS
COLLEGE OF SCIENCE
University of the Philippines
Diliman, Quezon City 1101 Metro-Manila

Tel. No. (632) 920-9749

Fax. No. (632) 928-0296



LETTER OF ACCEPTANCE

June 22, 2026

To: **ERIC M. INOCENCIO, MSc.**
OJT-Coordinator – BS Applied Physics Program
Department of Physical Sciences and Mathematics
College of Arts and Sciences
University of the Philippines – Manila

Dear Prof. Inocencio:

This is to inform you that Mr. **Miel Gabriel Lagdan** is accepted and will serve as a student intern in our institution – the **National Institute of Physics**.

Name (Last name, First Name, M.I.)	Student Number
Lagdan, Miel Gabriel	2023-14932

The duration of the internship program will be from June 22, 2026 – July 31, 2026, based on the university calendar; or on any other dates negotiated towards the OJT coordinator. We are glad to have your student be of service in our institution. We hope this will further our relationship as we serve with compassion and excellence.

Sincerely,

ERIC A. GALAPON, PhD.
Professor
National Institute of Physics
College of Science
University of the Philippines - Diliman

3.3 Proposed Training and Design Schedule

3.4 Pictures of Daily Journal

3.5 List of References Read During the Training

MAIN PAPERS:

1. Bagunu, R. J. C., & Galapon, E. A. (2025). *Quantization of the Hamilton equations of motion*. *Physica Scripta*, 100(5), 055208. <https://doi.org/10.1088/1402-4896/adc047>
2. Domingo, H. B., & Galapon, E. A. (2015). *Generalized Weyl transform for operator ordering: Polynomial functions in phase space*. *Journal of Mathematical Physics*, 56(2). <https://doi.org/10.1063/1.4907561>
3. Domingo, H. B. (2016). *Algebra of Non-Weyl Operators and Quantization of Analytic and Non-Analytic Classical Functions*. [Unpublished dissertation]. University of the Philippines Diliman.

RESOURCES:

1. Olsen, P. A., Rennie, S. J., & Goel, V. (2012). *Efficient automatic differentiation of matrix functions*. In *Lecture Notes in Computational Science and Engineering* (pp. 71–81). https://doi.org/10.1007/978-3-642-30023-3_7
2. Muller, F. (1997). *The equivalence myth of quantum mechanics—Part I. Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 28(1), 35–61. [https://doi.org/10.1016/S1355-2198\(96\)00022-6](https://doi.org/10.1016/S1355-2198(96)00022-6)
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5. McCoy, N. H. (1932). *On the Function in Quantum Mechanics Which Corresponds to a Given Function in Classical Mechanics*. *Proceedings of the National Academy of Sciences*, 18(11), 674–676. <https://doi.org/10.1073/pnas.18.11.674>
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10. Praxmeyer, L., & Wódkiewicz, K. (2003). *Hydrogen atom in phase space: The Kirkwood–Rihaczek representation*. *Physical Review A*, 67(5). <https://doi.org/10.1103/PhysRevA.67.054502>
11. Kirkwood, J. (1933). *Quantum Statistics of Almost Classical Assemblies*. *Physical Review*, 44(1), 31–37. <https://doi.org/10.1103/PhysRev.44.31>
12. Fan, H., Yuan, H., Cai, G., & Jiang, N. (2011). *Operators’ ordering: from Weyl ordering to normal ordering*. *Science China Physics, Mechanics and Astronomy*, 54(8), 1394–1397. <https://doi.org/10.1007/s11433-011-4404-z>
13. Fan, H. (1992). *Weyl ordering quantum mechanical operators by virtue of the IWWP technique*. *Journal of Physics A: Mathematical and General*, 25(11), 3443–3447. <https://doi.org/10.1088/0305-4470/25/11/043>
14. Tang, X., Sun, Y., & Su, H. (2010). *Weyl ordering symbol method for studying Wigner function of the damping field*. *International Journal of Theoretical Physics*, 49(9), 2016–2027. <https://doi.org/10.1007/s10773-010-0385-3>
15. Cohen, L. (2012). *The Weyl Operator and its Generalization*. <https://doi.org/10.1007/978-3-0348-0294-9>
16. Qian, M., & Xu, P. (1988). *Generalized Weyl transformation*. *Acta Mathematicae Applicatae Sinica English Series*, 4(3), 275–288. <https://doi.org/10.1007/BF02006225>
17. De Gosson, M. A., & Luef, F. (2015). *Born–Jordan pseudodifferential calculus, BOPP operators and deformation quantization*. *Integral Equations and Operator Theory*, 84(4), 463–485. <https://doi.org/10.1007/s00020-015-2273-y>
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23. Bargellini, A. E., & Ritelli, D. (2026). *An elementary approach to Euler’s reflection formula and its role in the infinite product of the sine function and the Basel problem*. *Foundations*, 6(2), 14. <https://doi.org/10.3390/foundations6020014>
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3.6 Abstract of Proposed Thesis

Quantization Ambiguities in Entanglement-Enhanced Interferometric Phase Generators

Miel Gabriel Lagdan, Herbert B. Domingo

Measurements involving photons are classically limited by the shot-noise limit. Entanglement in nonlinear interferometers enables these systems to approach or surpass the Heisenberg limit. However, the quantization of phase generators is plagued by ambiguities arising from different operator-ordering prescriptions (e.g., Weyl, Born-Jordan, and simplest symmetrization). This thesis investigates these ambiguities by deriving and comparing the phase sensitivities of entanglement-enhanced interferometric phase generators constructed under various ordering rules. Implications for quantum metrology in nonlinear media are discussed.

3.7 Resume

Please see attached file on next page.

MIEL GABRIEL LAGDAN

Applied Physics Undergraduate | Theoretical Physics Interests

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PROFESSIONAL SUMMARY

An applied physics undergraduate with a strong foundation in mathematical and theoretical physics, complemented by experience in undergraduate-level instruction and research. Knowledgeable in computational tools such as MATLAB, R, and Python, with research interests in theoretical and mathematical physics, particularly quantum theory and particle physics.

EXPERIENCE

VOLUNTEER TUTOR

University of the Philippines Manila
Ermita, Manila, Philippines

October 2023 – Present
PART-TIME

- Provides support to 30+ undergraduate students seeking reinforcement in learning topics involving algebra, calculus, real analysis, theoretical mechanics, electromagnetism, and quantum mechanics.
- Produces numerous helpful resources, including sample examinations, that helped many students in honing their skills in solving problems involving analysis and physics.
- Conducts one-to-one or many-to-one tutorial sessions to the students at the University of the Philippines Manila free of charge and helps other tutors with assistance in procuring resources for their students.

TUTOR

Great Minds Tutorial and Review Center
Quezon City, Philippines

August 2025 – Present
PART-TIME

- Provides supplementary help to 10+ elementary and high school students in topics related to mathematics and science (specifically physics) and recommends studying regimens tailored to each student.
- Creates and procures helpful materials in pedagogy useful to the student and the tutoring community.
- Assists students in preparation for upcoming academic competitions, nationally and internationally.

RESEARCH EXPERIENCE

Energies of two-flavor neutrino oscillations in uniform mass distribution using Dirac equation

2025

Proceedings of the Samahang Pisika ng Pilipinas

Miel Gabriel Lagdan, Jourdyn Roland N. Garcia, Herbert B. Domingo

- Performed theoretical analysis of neutrino oscillations of particles of uniform masses using coupled Dirac equations.
- Used Elisabetta Sassaroli’s model (1999) of neutrino oscillations and extended it by applying small mass increments.
- Provided a framework for uniform mass distribution models, with recommendations for extensions for varying mass.

Assessment of the Effectiveness of “Thermika,” a Serious Game, in Teaching Thermochemistry

2024

Philippine Federation of Chemistry Societies, Inc.

Miel Gabriel Lagdan, Pauline Khyla D. Gabatin, Niño Sherwin Lumancas, Roma Balagtas-Arguelles, Toshi Akira Capintog

- Development and statistical analysis of a serious game’s effectiveness to 84 high school students studying thermochemistry.
- Results demonstrate statistically significant and effective application of intervention with statistical evidence.
- An online, playable version was uploaded to itch.io, where students are free to access the game to assist in learning.

EDUCATION

Bachelor of Science in Applied Physics (Health Physics Concentration)

University of the Philippines Manila

Expected Graduation: 2027
August 2023 – Present

- With relevant coursework in radiation physics, statistical methods, magnetic resonance imaging, and computational physics.
- Joined a research laboratory in the University of the Philippines Manila called “Laboratory for Applied Mathematical Physics.”

Secondary Education

Caloocan City Science High School

September 2021 – July 2023

SKILLS

Technical Skills:

Python (numerical analysis)
MATLAB & R (statistical analysis, simulations)
LaTeX (technical writing)
Arduino (C++) (hardware interfacing)

Analytical Skills:

Mathematical modeling of physical systems
Differential equation analysis (ODEs & PDEs)
Approximation techniques
Statistical techniques

Other Skills:

Communication
Teamwork
Adaptability
Leadership

3.8 OUTPUT: Analysis of Paper (Bagunu & Galapon)

Please see attached file on next page.

Quantization of the Hamilton Equations of Motion [1]

Ramon Bagunu, Eric Galapon (Derived by Miel Lagdan)

July 17, 2026

Ramon Bagunu and Eric Galapon's paper entitled "Quantization of the Hamiltonian equations of motion" was investigated. Initially, papers suggest that Born-Jordan ordering *should* be the only ordering rule that will satisfy the relation between Schrodinger's matrix mechanics and Heisenberg's picture of quantum mechanics. The paper tackles on the investigation on the different images in quantization and asks if it will satisfy the Hamilton equations of motion in the quantum regime. The results show that arbitrary quantization will satisfy the Hamilton equations of motion, leading to a conclusion that all possible quantum images of the Hamiltonian will still exhibit the same quantum dynamical behavior and will display the correct quantum image.

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1 Introduction

It has been of interest in the field on how varying images on quantization affect measurement when acted upon. This paper, produced by Bagunu and Galapon, seek answers on how the Hamilton Equations of Motion are quantized using the different quantization images and see if there is a significant difference among them.

1.1 The Quantization Images

Suppose that for a Hamiltonian with polynomial terms $H(q, p) = p^m q^n$, the quantized image is given by [1, 4, 2] $\mathbf{H}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{m, n}$.

1. The **Weyl quantization** considers symmetric ordering such that

$$\mathbf{T}_{m,n} := \frac{1}{2^n} \sum_{j=0}^n \binom{n}{j} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} \quad \text{or} \quad \mathbf{T}_{m,n} = \frac{1}{2^m} \sum_{j=0}^m \binom{m}{j} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j} \quad (1)$$

2. The **simplest symmetrization** only considers equally satisfactory quantizations such that

$$\mathbf{S}_{m,n} := \frac{1}{2} (\mathbf{p}^m \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^m) \quad (2)$$

3. The **Born-Jordan** considers all terms to be equally weighted such that

$$\mathbf{B}_{m,n} := \frac{1}{n+1} \sum_{j=0}^n \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} \quad \text{or} \quad \mathbf{B}_{m,n} = \frac{1}{m+1} \sum_{j=0}^m \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j} \quad (3)$$

1.2 The Schrodinger and Heisenberg pictures

It is possible to relate the Schrodinger and Heisenberg pictures of quantum mechanics [3]. We start with such equation

$$\psi(t) = U(t, t_0) \psi(t_0) \quad (4)$$

provided that

$$U(t, t_0) = e^{iH_S(t-t_0)/\hbar} \quad (5)$$

is a family of unitary operators such that $i\hbar \frac{\partial \psi}{\partial t} = H_S \psi$. By for some observable A_S , the operators carry a dependence such that

$$A_H = U^\dagger A_S U \quad (6)$$

whereby taking the time derivative finds that

$$\frac{dA_H}{dt} = \left(\frac{i}{\hbar} U^\dagger H_S \right) A_S U + U^\dagger \left(\frac{\partial A_S}{\partial t} U \right) + U^\dagger A_S \left(-\frac{i}{\hbar} H_S U \right) \quad (7)$$

Simplifying, we get that

$$\frac{dA_H}{dt} = \frac{i}{\hbar} U^\dagger (H_S A_S - A_S H_S) U + U^\dagger \frac{\partial A_S}{\partial t} U \quad (8)$$

$$\frac{dA_H}{dt} = \frac{i}{\hbar} (H_H A_H - A_H H_H) + \frac{\partial A_H}{\partial t} \quad (9)$$

and finally

$$i\hbar \frac{dA_H}{dt} = i\hbar \frac{\partial A_H}{\partial t} + [A_H, H_H] \quad (10)$$

where H_H is the Hamiltonian in the Heisenberg picture. What follows is that suppose a ket in the Schrodinger picture is given

$$|\psi_S(t)\rangle = U(t, t_0) |\psi_S(t_0)\rangle \quad (11)$$

becomes the constant ket in the Heisenberg picture such that $|\psi_H\rangle = U(t, t_0)^\dagger |\psi_S(t)\rangle = |\psi(t_0)\rangle$ where the time-independence is removed. By the same formulation, we find that

$$|\psi_H\rangle = U(t, t_0)^\dagger |\psi_S(t)\rangle = |\psi_S(t_0)\rangle \quad (12)$$

Finally, it is found that

$$H_H(t) = U(t, t_0)^\dagger H_S U(t, t_0) \longrightarrow H_H(t) = H_S \quad (13)$$

where a few key points are added

- In the Heisenberg picture, the energy is constant.
- As the resulting energy is constant, in order for the equation (10) to be correct, it must be that the same quantization rules were applied.
- Heisenberg's matrix mechanics will only agree in the Schrodinger picture if the Born-Jordan quantization was used.

2 The Born-Jordan Quantization

The Born-Jordan Quantization. The derivation is as follows. We introduce two differentiation techniques [4] such that for the operator

$$\mathbf{y} = \prod_{m=1}^s \mathbf{y}_{l_m} = \mathbf{y}_{l_1} \mathbf{y}_{l_2} \mathbf{y}_{l_3} \cdots \mathbf{y}_{l_s} \quad (14)$$

the partial derivative with respect to $\mathbf{y}_k$ is provided by

$$\left(\frac{d\mathbf{y}}{d\mathbf{y}_k} \right)_1 = \sum_{r=1}^s \delta_{l,k} \prod_{m=r+1}^s \mathbf{y}_{l_m} \prod_{m=1}^{r-1} \mathbf{y}_{l_m} \quad (15)$$

$$\left(\frac{d\mathbf{y}}{d\mathbf{y}_k} \right)_2 = \lim_{\delta \rightarrow 0} \frac{\mathbf{y}(\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k + \delta \mathbf{I}, \dots, \mathbf{y}_s) - \mathbf{y}(\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_s)}{\delta} \quad (16)$$

where the resulting quantization must let the two differentiations be the same.

2.1 The partial derivative with respect to momentum

For an arbitrary Hamiltonian $H(q, p) = p^m q^n$,

$$\mathbf{H} = \sum_{j=0}^n a_j^{(n)} \mathbf{q}^{n-j} \mathbf{p}^m \mathbf{q}^j = \sum_{j=0}^m b_j^{(m)} \mathbf{p}^{m-j} \mathbf{q}^n \mathbf{p}^j \quad (17)$$

Expanding, we find

$$\mathbf{H} = a_0^{(n)} \mathbf{p}^m \mathbf{q}^n + a_1^{(n)} \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1} + a_2^{(n)} \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2} + \dots \quad (18)$$

and find the derivative of each term using two definitions of the partial derivative. For the first definition, we have the following:

$$\left(\frac{\partial \mathbf{p}^m \mathbf{q}^n}{\partial \mathbf{p}} \right)_1 = \mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \mathbf{p}^{m-3} \mathbf{q}^n \mathbf{p}^2 + \dots + \mathbf{q}^n \mathbf{p}^{m-1} \quad (19)$$

$$\left(\frac{\partial \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1}}{\partial \mathbf{p}} \right)_1 = \mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \mathbf{p}^{m-3} \mathbf{q}^n \mathbf{p}^2 + \dots + \mathbf{q}^n \mathbf{p}^{m-1} \quad (20)$$

$$\left(\frac{\partial \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2}}{\partial \mathbf{p}} \right)_1 = \mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \mathbf{p}^{m-3} \mathbf{q}^n \mathbf{p}^2 + \dots + \mathbf{q}^n \mathbf{p}^{m-1} \quad (21)$$

for which the same pattern emerges for all terms. Thus, we can generalize this formulation such that

$$\left(\frac{\partial \mathbf{H}}{\partial \mathbf{p}} \right)_1 = \sum_{k=0}^n a_k^{(n)} \sum_{j=0}^{m-1} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} \quad (22)$$

where the constant $\sum_{k=0}^n a_k^{(n)=1}$ is normalized. Thus

$$\left(\frac{\partial \mathbf{H}}{\partial \mathbf{p}} \right)_1 = \sum_{j=0}^{m-1} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} \quad (23)$$

We now find the derivative of $\mathbf{H}$ using the second definition. Allow,

$$\left(\frac{\partial \mathbf{p}^m \mathbf{q}^n}{\partial \mathbf{p}} \right)_2 = \lim_{\delta \rightarrow 0} \frac{(\mathbf{p} + \delta \mathbf{I})^m \mathbf{q}^n - \mathbf{p}^m \mathbf{q}^n}{\delta} = \lim_{\delta \rightarrow 0} \frac{\left(\sum_{k=0}^m \binom{m}{k} \mathbf{p}^{m-k} (\delta \mathbf{I})^k \right) \mathbf{q}^n - \mathbf{p}^m \mathbf{q}^n}{\delta} \quad (24)$$

$$= \lim_{\delta \rightarrow 0} \frac{\left(\sum_{k=1}^m \binom{m}{k} \mathbf{p}^{m-k} (\delta \mathbf{I})^k \right) \mathbf{q}^n}{\delta} = \lim_{\delta \rightarrow 0} m \mathbf{p}^{m-1} \mathbf{q}^n = m \mathbf{p}^{m-1} \mathbf{q}^n \quad (25)$$

$$\left(\frac{\partial \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1}}{\partial \mathbf{p}} \right)_2 = \lim_{\delta \rightarrow 0} \frac{\mathbf{q} (\mathbf{p} + \delta \mathbf{I})^m \mathbf{q}^{n-1} - \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1}}{\delta} = \lim_{\delta \rightarrow 0} \frac{\mathbf{q} \left(\sum_{k=0}^m \binom{m}{k} \mathbf{p}^{m-k} (\delta \mathbf{I})^k \right) \mathbf{q}^{n-1} - \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1}}{\delta} \quad (26)$$

$$= \lim_{\delta \rightarrow 0} \frac{\mathbf{q} \left(\sum_{k=1}^m \binom{m}{k} \mathbf{p}^{m-k} (\delta \mathbf{I})^k \right) \mathbf{q}^{n-1}}{\delta} = \lim_{\delta \rightarrow 0} m \mathbf{q} \mathbf{p}^{m-1} \mathbf{q}^{n-1} = m \mathbf{q} \mathbf{p}^{m-1} \mathbf{q}^{n-1} \quad (27)$$

$$\left(\frac{\partial \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2}}{\partial \mathbf{p}} \right)_2 = \lim_{\delta \rightarrow 0} \frac{\mathbf{q}^2 (\mathbf{p} + \delta \mathbf{I})^m \mathbf{q}^{n-2} - \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2}}{\delta} = \lim_{\delta \rightarrow 0} \frac{\mathbf{q}^2 \left(\sum_{k=0}^m \binom{m}{k} \mathbf{p}^{m-k} (\delta \mathbf{I})^k \right) \mathbf{q}^{n-2} - \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2}}{\delta} \quad (28)$$

$$= \lim_{\delta \rightarrow 0} \frac{\mathbf{q}^2 \left(\sum_{k=1}^m \binom{m}{k} \mathbf{p}^{m-k} (\delta \mathbf{I})^k \right) \mathbf{q}^{n-2}}{\delta} = \lim_{\delta \rightarrow 0} m \mathbf{q}^2 \mathbf{p}^{m-1} \mathbf{q}^{n-2} = m \mathbf{q}^2 \mathbf{p}^{m-1} \mathbf{q}^{n-2} \quad (29)$$

Thus, we can generalize this formulation such that

$$\left(\frac{\partial \mathbf{H}}{\partial \mathbf{p}} \right)_2 = m \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^{m-1} \mathbf{q}^{n-j} \quad (30)$$

and equating the two differentials together, we find

$$\left(\frac{\partial \mathbf{H}}{\partial \mathbf{p}} \right)_1 = \left(\frac{\partial \mathbf{H}}{\partial \mathbf{p}} \right)_2 \quad (31)$$

$$\sum_{j=0}^{m-1} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} = m \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^{m-1} \mathbf{q}^{n-j} \quad (32)$$

Recall that the Born-Jordan image is found using

$$\mathbf{B}_{m,n} := \frac{1}{n+1} \sum_{j=0}^n \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} = \frac{1}{m+1} \sum_{j=0}^m \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j} \quad (33)$$

$$\mathbf{B}_{m-1,n} := \frac{1}{n+1} \sum_{j=0}^n \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} = \frac{1}{m} \sum_{j=0}^m \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} \quad (34)$$

using 34, we rewrite 35,

$$\frac{m}{n+1} \sum_{j=0}^n \mathbf{q}^j \mathbf{p}^{m-1} \mathbf{q}^{n-j} = m \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^{m-1} \mathbf{q}^{n-j} \quad (35)$$

It must be that $a_j^{(n)} = \frac{1}{n+1}$. The Hamiltonian $\mathbf{H}$ should be written

$$\mathbf{H} = \frac{1}{n+1} \sum_{j=0}^n \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} \quad (36)$$

2.2 The partial derivative with respect to position

We similarly do the operations for the partial derivative with respect to the position $\mathbf{q}$. Expanding, we find

$$\mathbf{H} = b_0^{(n)} \mathbf{q}^n \mathbf{p}^m + b_1^{(n)} \mathbf{p} \mathbf{q}^n \mathbf{p}^{m-1} + b_2^{(n)} \mathbf{p}^2 \mathbf{q}^m \mathbf{q}^{n-2} + \dots \quad (37)$$

and find the derivative of each term using two definitions of the partial derivative. For the first definition, we have the following:

$$\left(\frac{\partial \mathbf{q}^m \mathbf{p}^n}{\partial \mathbf{q}} \right)_1 = \mathbf{q}^{n-1} \mathbf{p} + \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \mathbf{q}^{n-3} \mathbf{p}^m \mathbf{q}^2 + \dots + \mathbf{p}^m \mathbf{q}^{n-1} \quad (38)$$

$$\left(\frac{\partial \mathbf{p} \mathbf{q}^n \mathbf{p}^{m-1}}{\partial \mathbf{q}} \right)_1 = \mathbf{q}^{n-1} \mathbf{p} + \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \mathbf{q}^{n-3} \mathbf{p}^m \mathbf{q}^2 + \dots + \mathbf{p}^m \mathbf{q}^{n-1} \quad (39)$$

$$\left(\frac{\partial \mathbf{p}^2 \mathbf{q}^n \mathbf{p}^{m-2}}{\partial \mathbf{p}} \right)_1 = \mathbf{q}^{n-1} \mathbf{p} + \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \mathbf{q}^{n-3} \mathbf{p}^m \mathbf{q}^2 + \dots + \mathbf{p}^m \mathbf{q}^{n-1} \quad (40)$$

for which the same pattern emerges for all terms. Thus, we can generalize this formulation such that

$$\left(\frac{\partial \mathbf{H}}{\partial \mathbf{q}}\right)_1 = \sum_{k=0}^m b_k^{(m)} \sum_{j=0}^{n-1} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} \quad (41)$$

where the constant $\sum_{k=0}^n b_k^{(m)=1}$ is normalized. Thus

$$\left(\frac{\partial \mathbf{H}}{\partial \mathbf{q}}\right)_1 = \sum_{j=0}^{n-1} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} \quad (42)$$

We now find the derivative of $\mathbf{H}$ using the second definition. Allow,

$$\left(\frac{\partial \mathbf{q}^n \mathbf{p}^m}{\partial \mathbf{q}}\right)_2 = \lim_{\delta \rightarrow 0} \frac{(\mathbf{q} + \delta \mathbf{I})^n \mathbf{p}^m - \mathbf{q}^n \mathbf{p}^m}{\delta} = \lim_{\delta \rightarrow 0} \frac{\left(\sum_{k=0}^n \binom{n}{k} \mathbf{q}^{n-k} (\delta \mathbf{I})^k\right) \mathbf{p}^m - \mathbf{q}^n \mathbf{p}^m}{\delta} \quad (43)$$

$$= \lim_{\delta \rightarrow 0} \frac{\left(\sum_{k=1}^n \binom{n}{k} \mathbf{q}^{n-k} (\delta \mathbf{I})^k\right) \mathbf{p}^m}{\delta} = \lim_{\delta \rightarrow 0} n \mathbf{q}^{n-1} \mathbf{p}^m = n \mathbf{q}^{n-1} \mathbf{p}^m \quad (44)$$

$$\left(\frac{\partial \mathbf{p} \mathbf{q}^n \mathbf{p}^{m-1}}{\partial \mathbf{q}}\right)_2 = \lim_{\delta \rightarrow 0} \frac{\mathbf{p}(\mathbf{q} + \delta \mathbf{I})^n \mathbf{p}^{m-1} - \mathbf{p} \mathbf{q}^n \mathbf{p}^{m-1}}{\delta} = \lim_{\delta \rightarrow 0} \frac{\mathbf{p} \left(\sum_{k=0}^n \binom{n}{k} \mathbf{q}^{n-k} (\delta \mathbf{I})^k\right) \mathbf{p}^{m-1} - \mathbf{p} \mathbf{q}^n \mathbf{p}^{m-1}}{\delta} \quad (45)$$

$$= \lim_{\delta \rightarrow 0} \frac{\mathbf{p} \left(\sum_{k=1}^n \binom{n}{k} \mathbf{q}^{n-k} (\delta \mathbf{I})^k\right) \mathbf{p}^{m-1}}{\delta} = \lim_{\delta \rightarrow 0} n \mathbf{p} \mathbf{q}^{n-1} \mathbf{p}^{m-1} = n \mathbf{p} \mathbf{q}^{n-1} \mathbf{p}^{m-1} \quad (46)$$

$$\left(\frac{\partial \mathbf{p}^2 \mathbf{q}^n \mathbf{p}^{m-2}}{\partial \mathbf{q}}\right)_2 = \lim_{\delta \rightarrow 0} \frac{\mathbf{p}^2 (\mathbf{q} + \delta \mathbf{I})^n \mathbf{p}^{m-2} - \mathbf{p}^2 \mathbf{q}^n \mathbf{p}^{m-2}}{\delta} = \lim_{\delta \rightarrow 0} \frac{\mathbf{p}^2 \left(\sum_{k=0}^n \binom{n}{k} \mathbf{q}^{n-k} (\delta \mathbf{I})^k\right) \mathbf{p}^{m-2} - \mathbf{p}^2 \mathbf{q}^n \mathbf{p}^{m-2}}{\delta} \quad (47)$$

$$= \lim_{\delta \rightarrow 0} \frac{\mathbf{p}^2 \left(\sum_{k=1}^n \binom{n}{k} \mathbf{q}^{n-k} (\delta \mathbf{I})^k\right) \mathbf{p}^{m-2}}{\delta} = \lim_{\delta \rightarrow 0} n \mathbf{p}^2 \mathbf{q}^{n-1} \mathbf{p}^{m-2} = n \mathbf{p}^2 \mathbf{q}^{n-1} \mathbf{p}^{m-2} \quad (48)$$

Thus, we can generalize this formulation such that

$$\left(\frac{\partial \mathbf{H}}{\partial \mathbf{q}}\right)_2 = n \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^{n-1} \mathbf{p}^{m-j} \quad (49)$$

and equating the two differentials together, we find

$$\left(\frac{\partial \mathbf{H}}{\partial \mathbf{q}}\right)_1 = \left(\frac{\partial \mathbf{H}}{\partial \mathbf{q}}\right)_2 \quad (50)$$

$$\sum_{j=0}^{n-1} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} = n \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^{n-1} \mathbf{p}^{m-j} \quad (51)$$

Recall that the Born-Jordan image is found using

$$\mathbf{B}_{m,n} := \frac{1}{n+1} \sum_{j=0}^n \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} = \frac{1}{m+1} \sum_{j=0}^m \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j} \quad (52)$$

$$\mathbf{B}_{m,n-1} := \frac{1}{n} \sum_{j=0}^n \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} = \frac{1}{m+1} \sum_{j=0}^m \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j} \quad (53)$$

using 53, we rewrite 51,

$$\frac{n}{m+1} \sum_{j=0}^m \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j} = n \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^{n-1} \mathbf{p}^{m-j} \quad (54)$$

It must be that $b_j^{(m)} = \frac{1}{m+1}$. The Hamiltonian $\mathbf{H}$ should be written

$$\mathbf{H} = \frac{1}{m+1} \sum_{j=0}^m \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j} \quad (55)$$

We can see therefore that the partial derivatives are only variations of the Born-Jordan images, that is

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_1 = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_2 = m \mathbf{B}_{m-1,n} \quad (56)$$

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}} \right)_1 = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}} \right)_2 = n \mathbf{B}_{m,n-1} \quad (57)$$

3 The Differential Quotient of First-Type

We consider the differential quotient of first-type for images of the Weyl and simplest symmetrization.

3.1 Weyl-generated images

We find the partial derivative with respect to the momentum. From the definition of the Weyl image from 1

$$\mathbf{T}_{m,n} = \frac{1}{2^n} \sum_{j=0}^n \binom{n}{j} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} = \frac{1}{2^n} \left(\binom{n}{0} \mathbf{p}^m \mathbf{q}^n + \binom{n}{1} \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1} + \binom{n}{2} \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2} + \dots + \binom{n}{n} \mathbf{q}^n \mathbf{p}^m \right) \quad (58)$$

where we find the images of each of the terms to be

$$\left(\frac{\partial \mathbf{p}^m \mathbf{q}^n}{\partial \mathbf{p}} \right)_1 = \mathbf{p}^{m-1} \mathbf{q} + \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \mathbf{p}^{m-3} \mathbf{q}^n \mathbf{p}^2 + \dots + \mathbf{q}^n \mathbf{p}^{m-1} \quad (59)$$

$$\left(\frac{\partial \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1}}{\partial \mathbf{p}} \right)_1 = \mathbf{p}^{m-1} \mathbf{q} + \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \mathbf{p}^{m-3} \mathbf{q}^n \mathbf{p}^2 + \dots + \mathbf{q}^n \mathbf{p}^{m-1} \quad (60)$$

$$\left(\frac{\partial \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2}}{\partial \mathbf{p}} \right)_1 = \mathbf{p}^{m-1} \mathbf{q} + \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \mathbf{p}^{m-3} \mathbf{q}^n \mathbf{p}^2 + \dots + \mathbf{q}^n \mathbf{p}^{m-1} \quad (61)$$

where we can generalize such that

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_1 = \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} \sum_{j=0}^{m-1} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} \quad (62)$$

but $\sum_{k=0}^n \binom{n}{k} = 2^n$. Thus

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_1 = \sum_{j=0}^{m-1} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} = m \mathbf{B}_{m-1,n} \quad (63)$$

Similarly, for the partial derivative with respect to the position, we define the terms to be

$$\mathbf{T}_{m,n} = \frac{1}{2^m} \sum_{j=0}^m \binom{m}{j} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j} = \frac{1}{2^m} \left(\binom{m}{0} \mathbf{q}^n \mathbf{p}^m + \binom{m}{1} \mathbf{p} \mathbf{q}^n \mathbf{p}^{m-1} + \binom{m}{2} \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2} + \dots + \binom{m}{m} \mathbf{p}^m \mathbf{q}^n \right) \quad (64)$$

where we find the images of each of the terms to be

$$\left(\frac{\partial \mathbf{q}^m \mathbf{p}^n}{\partial \mathbf{q}} \right)_1 = \mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \mathbf{q}^{n-3} \mathbf{p}^m \mathbf{q}^2 + \dots + \mathbf{p}^m \mathbf{q}^{n-1} \quad (65)$$

$$\left(\frac{\partial \mathbf{p} \mathbf{q}^n \mathbf{p}^{m-1}}{\partial \mathbf{q}} \right)_1 = \mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \mathbf{q}^{n-3} \mathbf{p}^m \mathbf{q}^2 + \dots + \mathbf{p}^m \mathbf{q}^{n-1} \quad (66)$$

$$\left(\frac{\partial \mathbf{p}^2 \mathbf{q}^n \mathbf{p}^{m-2}}{\partial \mathbf{q}} \right)_1 = \mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \mathbf{q}^{n-3} \mathbf{p}^m \mathbf{q}^2 + \dots + \mathbf{p}^m \mathbf{q}^{n-1} \quad (67)$$

where we can generalize such that

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}}\right)_1 = \frac{1}{2^m} \sum_{k=0}^m \binom{m}{k} \sum_{j=0}^{n-1} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} \quad (68)$$

but $\sum_{k=0}^m \binom{m}{k} = 2^m$. Thus

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}}\right)_1 = \sum_{j=0}^{n-1} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} = n \mathbf{B}_{m,n-1} \quad (69)$$

3.2 Simplest symmetrization generated terms

Consider the generated images from simplest symmetrization

$$\mathbf{S}_{m,n} := \frac{1}{2} (\mathbf{p}^m \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^m) \quad (70)$$

where we find the partial derivative of each term to be

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}}\right)_1 = \frac{1}{2} (2[\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \mathbf{p}^{m-3} \mathbf{q}^n \mathbf{p}^2 + \cdots + \mathbf{q}^n \mathbf{p}^{m-1}]) = m \mathbf{B}_{m-1,n} \quad (71)$$

Similarly, for the partial derivative with respect to the position, we define

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}}\right)_1 = \frac{1}{2} (2[\mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \mathbf{q}^{n-3} \mathbf{p}^m \mathbf{q}^2 + \cdots + \mathbf{p}^m \mathbf{q}^{n-1}]) = n \mathbf{B}_{m,n-1} \quad (72)$$

We see, from the given derivations, the differential quotient of first type leads us to the Born-Jordan images with index shift $m-1$ and $n-1$ for differentials of momentum and position, respectively.

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}}\right)_1 = \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}}\right)_1 = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}}\right)_1 = m \mathbf{B}_{m-1,n} \quad (73)$$

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}}\right)_1 = \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}}\right)_1 = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}}\right)_1 = n \mathbf{B}_{m,n-1} \quad (74)$$

4 Modifications to the Differential Quotient of First Type

We now consider modifications to the formulation of the differential quotient of first type.

4.1 Differential quotient with Weyl basis

Allow the modification such that

$$\left(\frac{\partial \mathbf{y}}{\partial \mathbf{y}_k}\right)_W = \frac{n}{2^{n-1}} \sum_{r=1}^s \delta_{l_r, k} \binom{n-1}{l-1} \prod_{m=r+1}^s \mathbf{y}_{l_m} \prod_{m=1}^{r-1} \mathbf{y}_{l_m} \quad (75)$$

We use this to calculate for the partial derivative with respect to momentum and position for the quantization images mentioned here. Starting with the Weyl images, we have

$$\mathbf{T}_{m,n} = \frac{1}{2^n} \sum_{j=0}^n \binom{n}{j} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} = \frac{1}{2^n} \left(\binom{n}{0} \mathbf{p}^m \mathbf{q}^n + \binom{n}{1} \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1} + \binom{n}{2} \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2} + \cdots + \binom{n}{n} \mathbf{q}^n \mathbf{p}^m \right) \quad (76)$$

and finding the image's derivative, we find

$$\begin{aligned} \left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}}\right)_W &= \frac{1}{2^n} \frac{m}{2^{m-1}} \left(\binom{n}{0} \left[\binom{m-1}{0} \mathbf{p}^{m-1} \mathbf{q}^n + \binom{m-1}{1} \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \cdots + \binom{m-1}{m-1} \mathbf{q}^n \mathbf{p}^{m-1} \right] \right. \\ &\quad \left. + \binom{n}{1} \left[\binom{m-1}{0} \mathbf{p}^{m-1} \mathbf{q}^n + \binom{m-1}{1} \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \cdots + \binom{m-1}{m-1} \mathbf{q}^n \mathbf{p}^{m-1} \right] + \cdots \right) \end{aligned}$$

$$+ \binom{n}{n} \left[\binom{m-1}{0} \mathbf{p}^{m-1} \mathbf{q}^n + \binom{m-1}{1} \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \cdots + \binom{m-1}{m-1} \mathbf{q}^n \mathbf{p}^{m-1} \right] \quad (77)$$

which simplifies to

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_W = \frac{1}{2^n} \frac{m}{2^{m-1}} \sum_{k=0}^n \binom{n}{k} \sum_{j=0}^{m-1} \binom{m-1}{j} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} \quad (78)$$

but $\sum_{k=0}^n \binom{n}{k} = 2^n$. Thus

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_W = \frac{m}{2^{m-1}} \sum_{j=0}^{m-1} \binom{m-1}{j} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} = m \mathbf{T}_{m-1,n} \quad (79)$$

Similarly, for the case of the partial derivative with respect to the position, we find that

$$\begin{aligned} \left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_W &= \frac{1}{2^m} \frac{n}{2^{n-1}} \left(\binom{m}{0} \left[\binom{n-1}{0} \mathbf{q}^{n-1} \mathbf{p}^m + \binom{n-1}{1} \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \cdots + \binom{n-1}{n-1} \mathbf{p}^m \mathbf{q}^{n-1} \right] \right. \\ &\quad + \binom{m}{1} \left[\binom{n-1}{0} \mathbf{q}^{n-1} \mathbf{p}^m + \binom{n-1}{1} \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \cdots + \binom{n-1}{n-1} \mathbf{p}^m \mathbf{q}^{n-1} \right] + \cdots \\ &\quad \left. + \binom{m}{m} \left[\binom{n-1}{0} \mathbf{q}^{n-1} \mathbf{p}^m + \binom{n-1}{1} \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \cdots + \binom{n-1}{n-1} \mathbf{p}^m \mathbf{q}^{n-1} \right] \right) \quad (80) \end{aligned}$$

which simplifies to

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_W = \frac{1}{2^m} \frac{n}{2^{n-1}} \sum_{k=0}^m \binom{m}{k} \sum_{j=0}^{n-1} \binom{n-1}{j} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} \quad (81)$$

but $\sum_{k=0}^m \binom{m}{k} = 2^m$. Thus

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_W = \frac{n}{2^{n-1}} \sum_{j=0}^{n-1} \binom{n-1}{j} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} = n \mathbf{T}_{m,n-1} \quad (82)$$

As for the simplest symmetrization images, we have

$$\mathbf{S}_{m,n} := \frac{1}{2} (\mathbf{p}^m \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^m) \quad (83)$$

and finding the image's derivative, we find

$$\begin{aligned} \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}} \right)_W &= \frac{1}{2} \frac{m}{2^{m-1}} \left(\left[\binom{m-1}{0} \mathbf{p}^{m-1} \mathbf{q}^n + \binom{m-1}{1} \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \cdots + \binom{m-1}{m-1} \mathbf{q}^n \mathbf{p}^{m-1} \right] \right. \\ &\quad \left. + \left[\binom{m-1}{0} \mathbf{p}^{m-1} \mathbf{q}^n + \binom{m-1}{1} \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \cdots + \binom{m-1}{m-1} \mathbf{q}^n \mathbf{p}^{m-1} \right] \right) \quad (84) \end{aligned}$$

which simplifies to

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}} \right)_W = \frac{m}{2^{m-1}} \sum_{j=0}^{m-1} \binom{m-1}{j} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} = m \mathbf{T}_{m-1,n} \quad (85)$$

Similarly, for the case of the partial derivative with respect to the position, we find that

$$\begin{aligned} \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}} \right)_W &= \frac{1}{2} \frac{n}{2^{n-1}} \left(\left[\binom{n-1}{0} \mathbf{q}^{n-1} \mathbf{p}^m + \binom{n-1}{1} \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \cdots + \binom{n-1}{n-1} \mathbf{p}^m \mathbf{q}^{n-1} \right] \right. \\ &\quad \left. + \left[\binom{n-1}{0} \mathbf{q}^{n-1} \mathbf{p}^m + \binom{n-1}{1} \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \cdots + \binom{n-1}{n-1} \mathbf{p}^m \mathbf{q}^{n-1} \right] \right) \quad (86) \end{aligned}$$

which simplifies to

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}} \right)_W = \frac{n}{2^{n-1}} \sum_{j=0}^{n-1} \binom{n-1}{j} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} = n \mathbf{T}_{m,n-1} \quad (87)$$

For the Born-Jordan images, we find for the terms

$$\mathbf{B}_{m,n} := \frac{1}{n+1} \sum_{j=0}^n \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} = \frac{1}{n+1} (\mathbf{p}^m \mathbf{q}^n + \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1} + \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2} + \dots + \mathbf{q}^n \mathbf{p}^m) \quad (88)$$

and finding the image's derivative, we find

$$\begin{aligned} \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_W &= \frac{m}{2^{m-1}} \frac{1}{n+1} \left(\left[\binom{m-1}{0} \mathbf{p}^{m-1} \mathbf{q}^n + \binom{m-1}{1} \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \dots + \binom{m-1}{m-1} \mathbf{q}^n \mathbf{p}^{m-1} \right] \right. \\ &\quad + \left[\binom{m-1}{0} \mathbf{p}^{m-1} \mathbf{q}^n + \binom{m-1}{1} \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \dots + \binom{m-1}{m-1} \mathbf{q}^n \mathbf{p}^{m-1} \right] \\ &\quad \left. + \left[\binom{m-1}{0} \mathbf{p}^{m-1} \mathbf{q}^n + \binom{m-1}{1} \mathbf{p}^{m-2} \mathbf{q}^n \mathbf{p} + \dots + \binom{m-1}{m-1} \mathbf{q}^n \mathbf{p}^{m-1} \right] + \dots \right) \quad (89) \end{aligned}$$

which continues repeatedly for $n+1$ terms. That is, simplifying

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_W = \frac{m}{2^{m-1}} \frac{1}{n+1} \sum_{j=0}^{m-1} \binom{m-1}{j} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-1-j} (n+1) = m \mathbf{T}_{m-1,n} \quad (90)$$

Similarly, for the case of the partial derivative with respect to the position, we find that

$$\begin{aligned} \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}} \right)_W &= \frac{n}{2^{n-1}} \frac{1}{m+1} \left(\left[\binom{n-1}{0} \mathbf{q}^{n-1} \mathbf{p}^m + \binom{n-1}{1} \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \dots + \binom{n-1}{n-1} \mathbf{p}^m \mathbf{q}^{n-1} \right] \right. \\ &\quad + \left[\binom{n-1}{0} \mathbf{q}^{n-1} \mathbf{p}^m + \binom{n-1}{1} \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \dots + \binom{n-1}{n-1} \mathbf{p}^m \mathbf{q}^{n-1} \right] \\ &\quad \left. + \left[\binom{n-1}{0} \mathbf{q}^{n-1} \mathbf{p}^m + \binom{n-1}{1} \mathbf{q}^{n-2} \mathbf{p}^m \mathbf{q} + \dots + \binom{n-1}{n-1} \mathbf{p}^m \mathbf{q}^{n-1} \right] + \dots \right) \quad (91) \end{aligned}$$

which continues repeatedly for $m+1$ terms. That is, simplifying

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}} \right)_W = \frac{n}{2^{n-1}} \frac{1}{m+1} \sum_{j=0}^{n-1} \binom{n-1}{j} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-1-j} (m+1) = n \mathbf{T}_{m,n-1} \quad (92)$$

From this modified differential quotient, all of the results indicate that they are in the form of the Weyl images.

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_W = \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}} \right)_W = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_W = m \mathbf{T}_{m-1,n} \quad (93)$$

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_W = \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}} \right)_W = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}} \right)_W = n \mathbf{T}_{m,n-1} \quad (94)$$

4.2 Differential quotient with simplest symmetrization basis

Consider the modified formulation such that

$$\left(\frac{\partial \mathbf{y}}{\partial \mathbf{y}_k} \right)_S = \frac{n}{2} \sum_{r=1}^s \delta_{l_r,k} (\delta_{l,1} + \delta_{l,k}) \prod_{m=r+1}^s \mathbf{y}_{l_m} \prod_{m=1}^{m-1} \mathbf{y}_{l_m} \quad (95)$$

We use this to calculate for the partial derivative with respect to momentum and position for the quantization images mentioned here. Starting with the Weyl images, we have

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_S = \frac{m}{2} \frac{1}{2^n} \left(\binom{n}{0} [\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}] + \binom{n}{1} [\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}] + \dots + \binom{n}{m} [\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}] \right) \quad (96)$$

which simplifies to

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_S = \frac{m}{2^{n+1}} \sum_{k=0}^n \binom{n}{k} (\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}) \quad (97)$$

but $\sum_{k=0}^n \binom{n}{k} = 2^n$. Thus

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_S = \frac{m}{2} (\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}) = m \mathbf{S}_{m-1,n} \quad (98)$$

Similarly, for the case of the partial derivative with respect to the position, we find that

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_S = \frac{n}{2} \frac{1}{2^m} \left(\binom{m}{0} [\mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{p}^m \mathbf{q}^{n-1}] + \binom{m}{1} [\mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{p}^m \mathbf{q}^{n-1}] + \dots + \binom{m}{m} [\mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{p}^m \mathbf{q}^{n-1}] \right) \quad (99)$$

which simplifies to

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_S = \frac{n}{2^{m+1}} \sum_{k=0}^m \binom{m}{k} (\mathbf{p}^m \mathbf{q}^{n-1} + \mathbf{q}^{n-1} \mathbf{p}^m) \quad (100)$$

but $\sum_{k=0}^m \binom{m}{k} = 2^m$. Thus

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_S = \frac{n}{2} (\mathbf{p}^m \mathbf{q}^{n-1} + \mathbf{q}^{n-1} \mathbf{p}^m) = n \mathbf{S}_{m,n-1} \quad (101)$$

As for the simplest symmetrization images, we have

$$\mathbf{S}_{m,n} := \frac{1}{2} (\mathbf{p}^m \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^m) \quad (102)$$

and finding the image's derivative, we find

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}} \right)_S = \frac{m}{2} \frac{1}{2} ([\mathbf{p}^{m-1} \mathbf{q} + \mathbf{q}^n \mathbf{p}^{m-1}] + [\mathbf{p}^{m-1} \mathbf{q} + \mathbf{q}^n \mathbf{p}^{m-1}]) \quad (103)$$

which simplifies to

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}} \right)_S = \frac{m}{2} (\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}) = m \mathbf{S}_{m-1,n} \quad (104)$$

Similarly, for the case of the partial derivative with respect to the position, we find that

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}} \right)_S = \frac{n}{2} \frac{1}{2} ([\mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{p}^m \mathbf{q}^{n-1}] + [\mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{p}^m \mathbf{q}^{n-1}]) \quad (105)$$

which simplifies to

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}} \right)_S = \frac{n}{2} (\mathbf{p}^m \mathbf{q}^{n-1} + \mathbf{q}^{n-1} \mathbf{p}^m) = n \mathbf{S}_{m,n-1} \quad (106)$$

For the Born-Jordan images, we find for the terms

$$\mathbf{B}_{m,n} := \frac{1}{n+1} \sum_{j=0}^n \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} = \frac{1}{n+1} (\mathbf{p}^m \mathbf{q}^n + \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-1} + \mathbf{q}^2 \mathbf{p}^m \mathbf{q}^{n-2} + \dots + \mathbf{q}^n \mathbf{p}^m) \quad (107)$$

and finding the image's derivative, we find

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_S = \frac{m}{2} \frac{1}{n+1} ([\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}] + [\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}] + [\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}] + \dots) \quad (108)$$

which continues repeatedly for $n+1$ terms. That is, simplifying

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_S = \frac{m}{2} \frac{1}{n+1} (\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}) (n+1) = \frac{m}{2} (\mathbf{p}^{m-1} \mathbf{q}^n + \mathbf{q}^n \mathbf{p}^{m-1}) = m \mathbf{S}_{m-1,n} \quad (109)$$

Similarly, for the case of the partial derivative with respect to the position, we find that

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}} \right)_S = \frac{n}{2} \frac{1}{m+1} ([\mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{p}^m \mathbf{q}^{n-1}] + [\mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{p}^m \mathbf{q}^{n-1}] + [\mathbf{q}^{n-1} \mathbf{p}^m + \mathbf{p}^m \mathbf{q}^{n-1}] + \dots) \quad (110)$$

which continues repeatedly for $n + 1$ terms. That is, simplifying

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}}\right)_S = \frac{n}{2} \frac{1}{m+1} (\mathbf{p}^m \mathbf{q}^{n-1} + \mathbf{q}^{n-1} \mathbf{p}^m) (m+1) = \frac{n}{2} (\mathbf{p}^m \mathbf{q}^{n-1} + \mathbf{q}^{n-1} \mathbf{p}^m) = n \mathbf{S}_{m,n-1} \quad (111)$$

From this modified differential quotient, all of the results indicate that they are in the form of simplest symmetrization images.

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}}\right)_S = \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}}\right)_S = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}}\right)_S = m \mathbf{S}_{m-1,n} \quad (112)$$

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}}\right)_S = \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}}\right)_S = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}}\right)_S = n \mathbf{S}_{m,n-1} \quad (113)$$

Redefining the original differential quotient of first type to leads to a preferred quantization. It can be observed that these modified differentiation rules cannot be generalized into arbitrary ordering, and that they are tailored to be consistent only in their respective basis.

5 The Differential Quotient of the Second-Type

The generality of the differential quotient of the second-type is explored. This method was explored in a paper titled "General Formulation of Quantum Analysis,"[5] where quantum derivatives were developed for non-commutative operators in the Banach space, which utilizes the Leibniz rule.

5.1 Suzuki's Analysis

Suzuki's 1999 paper [5] starts with the definition of the Gâteaux differential and (G)-differentiability, which begins through an explanation of the Banach space.

Definition 5.1. A **Banach space** is a normed space which is complete in the metric defined by its norm. This means that every Cauchy sequence is required to converge.[6, 5, 7]

The importance of working in Banach spaces lies in their completeness. Because every Cauchy sequence converges, limiting processes, such as those used to define derivatives, are well-behaved and remain within the space. In particular, the norm induces a metric given by $d(x, y) := \|x - y\| = \|y - x\|$. This completeness is crucial when studying differentiability, as it guarantees that the relevant limits exist in the space. This allows for the following definition.

Definition 5.2. Let $y = f(x)$ on $\mathcal{X}$ to $\mathcal{D}$ be defined on an finitely open set $\mathcal{D}$ and suppose that for every $x \in \mathcal{D}$ and $h \in \mathcal{X}$ the quotient $[f(x + \zeta h) - f(x)]/\zeta$, which defined for $0 < |\zeta| < \rho(x, h)$, tends to the unique limit as $\zeta \rightarrow 0$. We then say that

1. $f(x)$ is (G)-differentiable in $\mathcal{D}$;
- 2.

$$\delta f(x; h) = \delta_x^h f = \lim_{\zeta \rightarrow 0} \frac{1}{\zeta} [f(x + \zeta h) - f(x)] \quad (114)$$

3. $f(x)$ possesses a Gâteaux differential in $\mathcal{D}$.

At this juncture, it is important to discuss the concept of *homogeneity*.

Definition 5.3. A function $f(x)$ defined on $\mathcal{X}$ to $\mathcal{D}$ is said to be homogeneous of degree n if

$$f(\alpha x) = \alpha^n f(x) \quad (115)$$

and such a function is called bounded if there exists an $M > 0$ such that

$$\|f(x)\| \leq M \|x\|^n \quad (116)$$

For the purposes of this discussion, we define the Fréchet space.

Definition 5.4. Let $y = f(x)$ on $\mathcal{X}$ to $\mathcal{D}$ be defined on an open set $\mathcal{D}$. We say that $f(x)$ is (F)-differentiable and possesses a total or a Fréchet differential on $\mathcal{D}$ if

1. $\delta(x; h)$ exists as a bounded homogeneous function of degree one in h ;

2.

$$\lim_{\|h\| \rightarrow 0} \frac{1}{\|h\|} \|f(x+h) - f(x) - \delta f(x; h)\| \quad (117)$$

And in fact, bullet 2 is actually implied by 1.

Theorem 5.5. $f(x)$ is (G) -differentiable in the finitely open set $\mathcal{D}$ if and only if for every $x \in \mathcal{D}$ and $h \in \mathcal{X}$, $f(x + \zeta h)$ is a holomorphic function of ζ when $|\zeta| < \rho(x, h)$.

With the necessary requisites, we define an arbitrary operator $\mathbf{A}$. Thus, the Gâteaux differential reads

$$d\mathbf{F}(\mathbf{A}) = \lim_{h \rightarrow 0} \frac{\mathbf{F}(\mathbf{A} + h d\mathbf{A}) - f(\mathbf{A})}{h} \quad (118)$$

where $h \in \mathbb{C}$ and for a fixed operator $d\mathbf{A}$. As the Gâteaux differential exists, the function $\mathbf{F}(\mathbf{A})$ is considered (G) -differentiable and exists as a bounded homogeneous function of degree one in h . Thus, the Fréchet differential reads

$$\lim_{B \rightarrow 0} \frac{\|\mathbf{F}(\mathbf{A} + \mathbf{B}) - \mathbf{F}(\mathbf{A}) - \mathcal{L}(\mathbf{B})\|}{\|\mathbf{B}\|} \quad (119)$$

where $\mathcal{L}$ is the Fréchet derivative. The differentiation $df(\mathbf{A}/d\mathbf{A})$ of $f(\mathbf{A})$ with respect to $\mathbf{A}$ in the analysis is represented by

$$\frac{df(\mathbf{A})}{d\mathbf{A}} = f_1(\mathbf{A}, \delta_{\mathbf{A}}) \rightarrow d\mathbf{F}(\mathbf{A}) = f_1(\mathbf{A}, \delta_{\mathbf{A}}) \cdot d\mathbf{A} \quad (120)$$

where $f_1(\mathbf{A}, \delta_{\mathbf{A}})$ is expressed in terms of the relevant operator $\mathbf{A}$ and the inner derivation $\delta_{\mathbf{A}}$. This derivative is referred to as a *hyperoperator*. This inner derivation allows $\delta_{\mathbf{A}} \cdot Q \equiv [\mathbf{A}, Q] = \mathbf{A}Q - Q\mathbf{A}$.

Theorem 5.6. When $\mathbf{F}(\mathbf{A}) \in \mathcal{D}_{\mathbf{A}}$, we have

$$\delta_{\mathbf{A}} d\mathbf{F}(\mathbf{A}) = \delta_{\mathbf{F}(\mathbf{A})} d\mathbf{A} \quad (121)$$

and consequently

$$d\mathbf{F}(\mathbf{A}) = \delta_{\mathbf{A} \rightarrow d\mathbf{A}} \mathbf{F}(\mathbf{A}) = \int_0^1 \mathbf{F}^{(1)}(\mathbf{A} - t\delta_{\mathbf{A}}) dt \cdot d\mathbf{A} = \frac{\delta_{\mathbf{F}(\mathbf{A})}}{\delta_{d\mathbf{A}}} \cdot d\mathbf{A} \quad (122)$$

for any choice of definitions of the differential $d\mathbf{F}(\mathbf{A})$.

With this, we can rewrite

$$d\mathbf{F}(\mathbf{A}) = \lim_{h \rightarrow 0} \frac{\mathbf{F}(\mathbf{A} + h d\mathbf{A}) - f(\mathbf{A})}{h} = \frac{d\mathbf{F}(\mathbf{A})}{d\mathbf{A}} \cdot d\mathbf{A} \quad (123)$$

which follows the following theorem.

Theorem 5.7 (Invariance of the Quantum Derivative). When $\mathbf{F}(\mathbf{A}) \in \mathcal{D}_{\mathbf{A}}$, the quantum derivative $d\mathbf{F}(\mathbf{A})/d\mathbf{A}$ is invariant for any choice of definitions of the differential $d\mathbf{F}(\mathbf{A})$ satisfying the Leibniz rule, and it is given by

$$\frac{d\mathbf{F}(\mathbf{A})}{d\mathbf{A}} = \frac{\delta_{\mathbf{F}(\mathbf{A})}}{\delta_{d\mathbf{A}}} = \int_0^1 \mathbf{F}^{(1)}(\mathbf{A} - t\delta_{\mathbf{A}}) dt \quad (124)$$

where it is clear that regardless of the definitions of the differential of $d\mathbf{F}(\mathbf{A})$, the ratio of the two hyperoperators remains the same. It proves the invariance of the quantum derivative, different ways of defining the differential lead to the same quantum derivative (as long as they obey the Leibniz rule).

Finally, we set so that auxiliary operator $\{\mathbf{H}_j\}$ satisfy the following

- $[\mathbf{H}_j, \mathbf{H}_k] = 0$ for $j \neq k$
- $[\mathbf{H}_j, \mathbf{A}_k] = 0$ for $j \neq k$
- $[\mathbf{H}_j, [\mathbf{H}_k, \mathbf{A}_k]] = 0$ for $j \neq k$
- $[\mathbf{H}_j, [\mathbf{H}_j, \mathbf{A}_j]] = 0$

where using these auxiliary operators, we introduce the following partial differential

$$d_j \mathbf{F} \equiv [\mathbf{H}_j, \mathbf{F}] \equiv \delta_{\mathbf{H}_j} \mathbf{F} \quad (125)$$

and thus

$$d\mathbf{F}(\mathbf{A}) = [\mathbf{H}, \mathbf{F}(\mathbf{A})] \quad (126)$$

5.2 The Theorems

As the mathematical foundation is established, we begin the derivation for the quantization using the differential quotient of second-type. Given the classical function $f = f(q, p)$, its partial derivative is given by

$$\frac{\partial f(q, p)}{\partial p} = \lim_{\delta \rightarrow 0} \frac{f(q, p + \delta) - f(q, p)}{\delta} \quad \frac{\partial f(q, p)}{\partial q} = \lim_{\delta \rightarrow 0} \frac{f(q + \delta, p) - f(q, p)}{\delta} \quad (127)$$

and as such, for an operator function $\mathbf{F} = \mathbf{F}(\mathbf{q}, \mathbf{p})$ the partial derivative are

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\mathbf{p}} = \lim_{\delta \rightarrow 0} \frac{\mathbf{F}(\mathbf{q}, \mathbf{p} + \delta \mathbf{I}) - \mathbf{F}(\mathbf{q}, \mathbf{p})}{\delta} \quad \frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\mathbf{q}} = \lim_{\delta \rightarrow 0} \frac{\mathbf{F}(\mathbf{q} + \delta \mathbf{I}, \mathbf{p}) - \mathbf{F}(\mathbf{q}, \mathbf{p})}{\delta} \quad (128)$$

where the following is considered.

Theorem 5.8. *The differential quotient of second type is linear.*

Proof. Consider two operator functions such that $\mathbf{F} = \mathbf{F}(\mathbf{q}, \mathbf{p})$ and $\mathbf{G} = \mathbf{G}(\mathbf{q}, \mathbf{p})$ and two scalars a and b . Thus

$$\frac{\partial(a\mathbf{F} + b\mathbf{G})}{\partial \mathbf{p}} = \lim_{\delta \rightarrow 0} \frac{a\mathbf{F}(\mathbf{q}, \mathbf{p} + \delta \mathbf{I}) + b\mathbf{G}(\mathbf{q}, \mathbf{p} + \delta \mathbf{I}) - (a\mathbf{F}(\mathbf{q}, \mathbf{p}) + b\mathbf{G}(\mathbf{q}, \mathbf{p}))}{\delta} \quad (129)$$

Simplifying

$$\frac{\partial(a\mathbf{F} + b\mathbf{G})}{\partial \mathbf{p}} = \lim_{\delta \rightarrow 0} \frac{a\mathbf{F}(\mathbf{q}, \mathbf{p} + \delta \mathbf{I}) - a\mathbf{F}(\mathbf{q}, \mathbf{p})}{\delta} + \lim_{\delta \rightarrow 0} \frac{b\mathbf{G}(\mathbf{q}, \mathbf{p} + \delta \mathbf{I}) - b\mathbf{G}(\mathbf{q}, \mathbf{p})}{\delta} \quad (130)$$

and finally

$$\frac{\partial(a\mathbf{F} + b\mathbf{G})}{\partial \mathbf{p}} = a \lim_{\delta \rightarrow 0} \frac{\mathbf{F}(\mathbf{q}, \mathbf{p} + \delta \mathbf{I}) - \mathbf{F}(\mathbf{q}, \mathbf{p})}{\delta} + b \lim_{\delta \rightarrow 0} \frac{\mathbf{G}(\mathbf{q}, \mathbf{p} + \delta \mathbf{I}) - \mathbf{G}(\mathbf{q}, \mathbf{p})}{\delta} = a \frac{\partial \mathbf{F}}{\partial \mathbf{p}} + b \frac{\partial \mathbf{G}}{\partial \mathbf{p}} \quad (131)$$

□

And without loss for generality,

Theorem 5.9. *Let $f(q, p) = p^m q^n$ with its arbitrarily-quantized quantum image $\mathbf{F}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{m,n}$. The partial derivative is given by*

$$\frac{\partial f(q, p)}{\partial p} = m p^{m-1} q^n \longrightarrow \frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = m \mathbf{A}_{m-1,n} \quad (132)$$

and

$$\frac{\partial f(q, p)}{\partial q} = n p^m q^{n-1} \longrightarrow \frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = n \mathbf{A}_{m,n-1} \quad (133)$$

Proof. Consider the operator function $\mathbf{F}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{m,n} = \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j}$. Per the limit definition,

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{\mathbf{q}^j (\mathbf{p} + \delta \mathbf{I})^m \mathbf{q}^{n-j} - \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j}}{\delta} \quad (134)$$

and we expand the binomial term

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^n a_j^{(n)} \frac{\mathbf{q}^j \left(\sum_{k=0}^m \binom{m}{k} \mathbf{p}^{m-k} (\delta \mathbf{I})^k \right) \mathbf{q}^{n-j} - \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j}}{\delta} = \lim_{\delta \rightarrow 0} \sum_{j=0}^n a_j^{(n)} \frac{\mathbf{q}^j \sum_{k=1}^m \binom{m}{k} \mathbf{p}^{m-k} \delta^k \mathbf{q}^{n-j}}{\delta} \quad (135)$$

and further simplifying¹

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = m \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^{m-1} \mathbf{q}^{n-j} = m \mathbf{A}_{m-1,n} \quad (136)$$

Next, consider that $\mathbf{F}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{m,n} = \sum_{j=0}^n b_j^{(m)} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j}$. Per the limit definition,

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{\mathbf{q}^j (\mathbf{p} + \delta \mathbf{I})^m \mathbf{q}^{n-j} - \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j}}{\delta} \quad (137)$$

¹At $k > 1$, we see that $\delta^k > \delta$, and proceeding terms $k > 1$ are equal to 0.

and we expand the binomial term

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^m b_j^{(m)} \frac{\mathbf{p}^j \left(\sum_{k=0}^n \binom{n}{k} \mathbf{q}^{n-k} (\delta \mathbf{I})^k \right) \mathbf{p}^{m-j} - \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j}}{\delta} = \lim_{\delta \rightarrow 0} \sum_{j=0}^m b_j^{(m)} \frac{\mathbf{p}^j \sum_{k=1}^n \binom{n}{k} \mathbf{q}^{n-k} \delta^k \mathbf{p}^{m-j}}{\delta} \quad (138)$$

and further simplifying²

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = n \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^{n-1} \mathbf{p}^{m-j} = n \mathbf{A}_{m,n-1} \quad (139)$$

□

For completeness, let's consider the problem of negative indices.

Theorem 5.10. *Let $f = f(q, p) = p^{-m} q^n$ with its arbitrarily-quantized quantum image $\mathbf{F}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{-m,n}$. The partial derivative with respect to the momentum is given by*

$$\frac{\partial f(q, p)}{\partial p} = -m p^{-m-1} q^n \longrightarrow \frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = -m \mathbf{A}_{-m-1,n} \quad (140)$$

Proof. Consider the image

$$\mathbf{F}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{-m,n} = \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j} \quad (141)$$

such that we can define the derivative

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{a_j^{(n)}}{\delta} \left[\mathbf{q}^j (\mathbf{p} + \delta \mathbf{I})^{-m} \mathbf{q}^{n-j} - \mathbf{q}^j \mathbf{p}^{-m} \mathbf{q}^{n-j} \right] = \lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{a_j^{(n)}}{\delta} \left[\mathbf{q}^j \{ (\mathbf{p} + \delta \mathbf{I})^{-m} - \mathbf{p}^{-m} \} \mathbf{q}^{n-j} \right] \quad (142)$$

and we factor out a term $\mathbf{p}^{-m} (\mathbf{p} + \delta \mathbf{I})^{-m}$

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{a_j^{(n)}}{\delta} \left[\mathbf{q}^j \{ (\mathbf{p} + \delta \mathbf{I})^m - \mathbf{p}^m \} (\mathbf{p}^{-m} (\mathbf{p} + \delta \mathbf{I})^{-m}) \mathbf{q}^{n-j} \right] \quad (143)$$

and we expand the binomial

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{a_j^{(n)}}{\delta} \left[\mathbf{q}^j \left\{ \left(\mathbf{p}^m - \sum_{k=0}^m \binom{m}{k} \mathbf{p}^{m-k} (\delta \mathbf{I})^k \right) \right\} (\mathbf{p}^{-m} (\mathbf{p} + \delta \mathbf{I})^{-m}) \mathbf{q}^{n-j} \right] \quad (144)$$

and produce the terms for $k = 0$ such that

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{a_j^{(n)}}{\delta} \left[\mathbf{q}^j \left\{ \left(- \sum_{k=1}^m \binom{m}{k} \mathbf{p}^{m-k} (\delta \mathbf{I})^k \right) \right\} (\mathbf{p}^{-m} (\mathbf{p} + \delta \mathbf{I})^{-m}) \mathbf{q}^{n-j} \right] \quad (145)$$

and produce the terms for $k = 1$ (and disregarding the rest as $\delta^k > \delta$), we find

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = -m \sum_{j=0}^n a_j^{(n)} \left[\mathbf{q}^j \mathbf{p}^{m-1} \mathbf{p}^{-m} \mathbf{p}^{-m} \mathbf{q}^{n-j} \right] = -m \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^{-m-1} \mathbf{q}^{n-j} = -m \mathbf{A}_{-m-1,n} \quad (146)$$

□

Theorem 5.11. *Let $f = f(q, p) = p^m q^{-n}$ with its arbitrarily-quantized quantum image $\mathbf{F}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{m,-n}$. The partial derivative with respect to the position is given by*

$$\frac{\partial f(q, p)}{\partial q} = -n p^m q^{-n-1} \longrightarrow \frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = -n \mathbf{A}_{m,-n-1} \quad (147)$$

²At $k > 1$, we see that $\delta^k > \delta$, and proceeding terms $k > 1$ are equal to 0.

Proof. Consider the image

$$\mathbf{F}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{m,-n} = \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j} \quad (148)$$

such that we can define the derivative

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^m \frac{b_j^{(m)}}{\delta} \left[\mathbf{p}^j (\mathbf{q} + \delta \mathbf{I})^{-n} \mathbf{p}^{m-j} - \mathbf{p}^j \mathbf{q}^{-n} \mathbf{p}^{m-j} \right] = \lim_{\delta \rightarrow 0} \sum_{j=0}^m \frac{b_j^{(m)}}{\delta} [\mathbf{p}^j \{ (\mathbf{q} + \delta \mathbf{I})^{-n} - \mathbf{q}^{-n} \} \mathbf{p}^{m-j}] \quad (149)$$

and we factor our a term $\mathbf{q}^{-n}(\mathbf{q} + \delta \mathbf{I})^{-n}$

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^m \frac{b_j^{(m)}}{\delta} [\mathbf{p}^j \{ (\mathbf{q} + \delta \mathbf{I})^n - \mathbf{q}^n \} (\mathbf{q}^{-n}(\mathbf{q} + \delta \mathbf{I})^{-n}) \mathbf{p}^{m-j}] \quad (150)$$

and we expand the binomial

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^m \frac{b_j^{(m)}}{\delta} \left[\mathbf{p}^j \left\{ \left(\mathbf{q}^n - \sum_{k=0}^m \binom{n}{k} \mathbf{q}^{n-k} (\delta \mathbf{I})^k \right) \right\} (\mathbf{q}^{-n}(\mathbf{q} + \delta \mathbf{I})^{-n}) \mathbf{p}^{m-j} \right] \quad (151)$$

and produce the terms for $k = 0$ such that

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = \lim_{\delta \rightarrow 0} \sum_{j=0}^m \frac{b_j^{(m)}}{\delta} \left[\mathbf{p}^j \left\{ \left(- \sum_{k=1}^m \binom{n}{k} \mathbf{q}^{n-k} (\delta \mathbf{I})^k \right) \right\} (\mathbf{q}^{-n}(\mathbf{q} + \delta \mathbf{I})^{-n}) \mathbf{p}^{m-j} \right] \quad (152)$$

and produce the terms for $k = 1$ (and disregarding the rest as $\delta^k > \delta$), we find

$$\frac{\partial \mathbf{F}(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = -n \sum_{j=0}^m b_j^{(m)} [\mathbf{p}^j \mathbf{q}^{n-1} \mathbf{q}^{-n} \mathbf{q}^{-n} \mathbf{p}^{m-j}] = -n \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^{-n-1} \mathbf{p}^{m-j} = -n \mathbf{A}_{m,-n-1} \quad (153)$$

□

This generalization allows us to say that for any arbitrary quantization $A_{m,n}$ that the partial derivative using the differential quotient of the second-type will be invariant. No specific quantization is preferred by the differential quotient of the second-type. Recall that the quantizations of the differential quotient of the first-type was found to be

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_W = m \mathbf{T}_{m-1,n} \quad \left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_W = n \mathbf{T}_{m,n-1} \quad (154)$$

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}} \right)_S = m \mathbf{S}_{m-1,n} \quad \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}} \right)_S = n \mathbf{S}_{m,n-1} \quad (155)$$

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_1 = m \mathbf{B}_{m-1,n} \quad \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}} \right)_1 = n \mathbf{B}_{m,n-1} \quad (156)$$

and by Theorem 5.9, the following is considered true

$$\left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_W = \left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{p}} \right)_2 \quad \left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_W = \left(\frac{\partial \mathbf{T}_{m,n}}{\partial \mathbf{q}} \right)_2 \quad (157)$$

$$\left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}} \right)_S = \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{p}} \right)_2 \quad \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}} \right)_S = \left(\frac{\partial \mathbf{S}_{m,n}}{\partial \mathbf{q}} \right)_2 \quad (158)$$

$$\left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_1 = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{p}} \right)_2 \quad \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}} \right)_1 = \left(\frac{\partial \mathbf{B}_{m,n}}{\partial \mathbf{q}} \right)_2 \quad (159)$$

Theorem 5.12. For an arbitrarily-quantized operator $\mathbf{F}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{m,n}$ multiple differentiation with respect to the momentum yields

$$\frac{\partial^s \mathbf{A}_{m,n}}{\partial \mathbf{p}^s} = \frac{m!}{(m-s)!} \mathbf{A}_{m-s,n} \quad (160)$$

where $s \in \mathbb{Z}_+$.

Proof. Using mathematical induction, we first assume that, from Theorem 5.9 that this theorem is applicable for $s = u$. Thus, the $s = u + 1$ th derivative is then

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{p}^{u+1}} = \frac{\partial}{\partial \mathbf{p}} \left(\frac{n!}{(m-s)!} \mathbf{A}_{m-s,n} \right) = \frac{m!}{(m-s)!} \lim_{\delta \rightarrow 0} \sum_{j=0}^{m-s} \frac{a_j^{(n)}}{\delta} [\mathbf{q}^j \{(\mathbf{p} + \delta \mathbf{I})^{m-s} - \mathbf{p}^{m-s}\} \mathbf{q}^{n-j}] \quad (161)$$

and expanding the binomial term allows

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{p}^{u+1}} = \frac{m!}{(m-s)!} \lim_{\delta \rightarrow 0} \sum_{j=0}^{m-s} \frac{a_j^{(n)}}{\delta} \left[\mathbf{q}^j \left\{ \left(\sum_{k=0}^{m-s} \binom{m-s}{k} \mathbf{p}^{m-s-k} (\delta \mathbf{I})^k \right) - \mathbf{p}^{m-s} \right\} \mathbf{q}^{n-j} \right] \quad (162)$$

where allowing the binomial term be $k = 0$ removes the term $-\mathbf{p}^{m-s}$.

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{p}^{u+1}} = \frac{m!}{(m-s)!} \lim_{\delta \rightarrow 0} \sum_{j=0}^{m-s} \frac{a_j^{(n)}}{\delta} \left[\mathbf{q}^j \left\{ \left(\sum_{k=1}^{m-s} \binom{m-s}{k} \mathbf{p}^{m-s-k} (\delta \mathbf{I})^k \right) \right\} \mathbf{q}^{n-j} \right] \quad (163)$$

Further exploration of $k = 1$ onwards removes all other terms altogether.

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{p}^{u+1}} = \frac{m!}{(m-s)!} (m-s) \sum_{j=0}^{m-s} a_j^{(n)} \mathbf{q}^j \mathbf{p}^{m-s-1} \mathbf{q}^{n-j} = \quad (164)$$

Finally,

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{p}^{u+1}} = \frac{m!}{(m-s-1)!} \sum_{j=0}^{m-s} a_j^{(n)} \mathbf{q}^j \mathbf{p}^{m-s-1} \mathbf{q}^{n-j} = \frac{m!}{(m-s-1)!} \mathbf{A}_{m-s-1,n} \quad (165)$$

Since it's applicable for all $s = u + 1$, it must be its applicable for all $s \in \mathbb{Z}_+$. $\square$

Theorem 5.13. For an arbitrarily-quantized operator $\mathbf{F}(\mathbf{q}, \mathbf{p}) = \mathbf{A}_{m,n}$ multiple differentiation with respect to the position yields

$$\frac{\partial^t \mathbf{A}_{m,n}}{\partial \mathbf{q}^t} = \frac{n!}{(n-t)!} \mathbf{A}_{m,n-t} \quad (166)$$

where $t \in \mathbb{Z}_+$.

Proof. Using mathematical induction, we first assume that, from Theorem 5.9 that this theorem is applicable for $t = u$. Thus, the $t = u + 1$ th derivative is then

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{q}^{u+1}} = \frac{\partial}{\partial \mathbf{q}} \left(\frac{n!}{(n-t)!} \mathbf{A}_{m,n-t} \right) = \frac{n!}{(n-t)!} \lim_{\delta \rightarrow 0} \sum_{j=0}^{n-t} \frac{b_j^{(m)}}{\delta} [\mathbf{p}^j \{(\mathbf{q} + \delta \mathbf{I})^{n-t} - \mathbf{q}^{n-t}\} \mathbf{p}^{m-j}] \quad (167)$$

and expanding the binomial term allows

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{q}^{u+1}} = \frac{n!}{(n-t)!} \lim_{\delta \rightarrow 0} \sum_{j=0}^{n-t} \frac{b_j^{(m)}}{\delta} \left[\mathbf{p}^j \left\{ \left(\sum_{k=0}^{n-t} \binom{n-t}{k} \mathbf{q}^{n-t-k} (\delta \mathbf{I})^k \right) - \mathbf{q}^{n-t} \right\} \mathbf{p}^{m-j} \right] \quad (168)$$

where allowing the binomial term be $k = 0$ removes the term $-\mathbf{q}^{n-t}$.

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{q}^{u+1}} = \frac{n!}{(n-t)!} \lim_{\delta \rightarrow 0} \sum_{j=0}^{n-t} \frac{b_j^{(m)}}{\delta} \left[\mathbf{p}^j \left\{ \left(\sum_{k=1}^{n-t} \binom{n-t}{k} \mathbf{q}^{n-t-k} (\delta \mathbf{I})^k \right) \right\} \mathbf{p}^{m-j} \right] \quad (169)$$

Further exploration of $k = 1$ onwards removes all other terms altogether.

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{q}^{u+1}} = \frac{n!}{(n-t)!} (n-t) \sum_{j=0}^{n-t} b_j^{(m)} \mathbf{p}^j \mathbf{q}^{n-t-1} \mathbf{p}^{m-j} = \quad (170)$$

Finally,

$$\frac{\partial^{u+1} \mathbf{A}_{m,n}}{\partial \mathbf{q}^{u+1}} = \frac{n!}{(n-t-1)!} \sum_{j=0}^{n-t} b_j^{(m)} \mathbf{p}^j \mathbf{q}^{n-t-1} \mathbf{p}^{m-j} = \frac{n!}{(n-t-1)!} \mathbf{A}_{m,n-t-1} \quad (171)$$

Since it's applicable for all $t = u + 1$, it must be its applicable for all $t \in \mathbb{Z}_+$. $\square$

A few key points are to mind.

- The theorems proven here are to provide a complete proof for the full compatibility of differentiation rules to the operator regime.
- Such theorems are supposed to be consistent with classical rules of differentiation, and such rigor is applied to do so.

So far, the results have been consistent. In terms of multiple differentiation, it is evident that it follows the standard lowering exponent operation seen in polynomials with degree n . We apply in the case wherein it is possible for the indices to be negative.

Theorem 5.14. *For an arbitrarily-quantized operator $\mathbf{A}_{-m,n}$, multiple differentiation with respect to the momentum gives*

$$\frac{\partial^s \mathbf{A}_{-m,n}}{\partial \mathbf{p}^s} = (-1)^s \frac{(m+s-1)!}{(m-1)!} \mathbf{A}_{-m,n} \quad (172)$$

Proof. Begin with the case such that we have

$$\frac{\partial^s \mathbf{A}_{-m,n}}{\partial \mathbf{p}^s} = \frac{(-m)!}{(-m-s)!} \mathbf{A}_{-m,n} \quad (173)$$

However, for such factorials, it is given by the gamma function, that is

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt \quad (174)$$

We begin using the introduction of the beta function, that is,

$$\mathrm{B}(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt \quad (175)$$

The gamma function and beta functions are connected by the following relation. Allow for the equations

$$\Gamma(x)\Gamma(y) = \left(\int_0^\infty e^{-u} u^{x-1} du \right) \left(\int_0^\infty e^{-v} v^{y-1} dv \right) = \int_0^\infty \int_0^\infty e^{-(u+v)} u^{x-1} v^{y-1} du dv \quad (176)$$

and allow $s = u + v$ and $v = s - u$

$$\Gamma(x)\Gamma(y) = \int_0^\infty \int_0^\infty e^{-s} u^{x-1} (s-u)^{y-1} du ds = \int_0^\infty \int_0^\infty e^{-s} u^{x-1} \left(1 - \frac{u}{s}\right)^{y-1} s^{y-1} du ds \quad (177)$$

and allow for $t = \frac{u}{s}$ such that $u = st$ and $du = s dt$

$$\Gamma(x)\Gamma(y) = \int_0^\infty \int_0^\infty e^{-s} (st)^{x-1} (1-t)^{y-1} s^y dt ds = \left(\int_0^\infty e^{-s} s^{x+y-1} ds \right) \left(\int_0^1 t^{x-1} (1-t)^{y-1} dt \right) \quad (178)$$

and see that

$$\Gamma(x)\Gamma(y) = \Gamma(x+y)\mathrm{B}(x, y) \longrightarrow \mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} \quad (179)$$

In the case for where $-z$ and $y = 1 + z$, we see that

$$\Gamma(-z)\Gamma(1+z) = \int_0^1 t^{-z-1} (1-t)^z dt = -\pi \csc(\pi z) \longrightarrow \Gamma(-z) = \frac{-\pi \csc \pi z}{\Gamma(1+z)} \quad (180)$$

and we can rewrite the ratio of the multiple differentiation with respect to the momentum

$$\frac{(-m)!}{(-m-s)!} = \lim_{z \rightarrow m} \frac{(-z)!}{(-z-s)!} = \lim_{z \rightarrow m} \frac{\frac{-\pi \csc \pi z}{\Gamma(1+z)}}{\frac{-\pi \csc \pi(z+s)}{\Gamma(1+z+s)}} = \lim_{z \rightarrow m} \frac{\sin(\pi(z+s))\Gamma(1+z+s)}{\sin(\pi z)\Gamma(1+z)} = (-1)^s \frac{(m+s)!}{m!} \quad (181)$$

by which the act of differentiation reduces the constant m by acting as if a ladder operator. Thus the constant becomes $(-1)^s((m+s-1)!/(m-1)!)$. By mathematical induction, allow our theorem to be true when $s = u$. Then, for $s = u + 1$,

$$\frac{\partial^{u+1} \mathbf{A}_{-m,n}}{\partial \mathbf{p}} = \frac{\partial}{\partial \mathbf{p}} \left((-1)^u \frac{(m+u-1)!}{(m-1)!} \mathbf{A}_{-m-u,n} \right) \quad (182)$$

and by taking the derivative

$$\frac{\partial^{u+1} \mathbf{A}_{-m,n}}{\partial \mathbf{p}} = (-1)^u \frac{(m+u-1)!}{(m-1)!} \lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{a_j^{(n)}}{\delta} \mathbf{q}^j [(\mathbf{p} + \delta \mathbf{I})^{-m-u} + \mathbf{p}^{-m-u}] \mathbf{q}^{n-j} \quad (183)$$

We turn our attention to the summation. We allow

$$\lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{a_j^{(n)}}{\delta} \mathbf{q}^j [(\mathbf{p} + \delta \mathbf{I})^{-m-u} + \mathbf{p}^{-m-u}] \mathbf{q}^{n-j} = \lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{a_j^{(n)}}{\delta} \mathbf{q}^j [\mathbf{p}^{m+u} - (\mathbf{p} + \delta \mathbf{I})^{m+u}] (\mathbf{p}^{-m-u} (\mathbf{p} + \delta \mathbf{I})^{-m-u}) \mathbf{q}^{n-j} \quad (184)$$

and expanding

$$\lim_{\delta \rightarrow 0} \sum_{j=0}^n \frac{a_j^{(n)}}{\delta} \mathbf{q}^j \left[\mathbf{p}^{m+u} - \sum_{k=0}^{m+u} \binom{m+u}{k} \mathbf{p}^{m+u-k} \delta^k \right] (\mathbf{p}^{-m-u} (\mathbf{p} + \delta \mathbf{I})^{-m-u}) \mathbf{q}^{n-j} \quad (185)$$

where we further see that

$$\sum_{j=0}^n a_j^{(n)} \mathbf{q}^j [-(m+u) \mathbf{p}^{m+u-1}] (\mathbf{p}^{-m-u} \mathbf{p}^{-m-u}) \mathbf{q}^{n-j} = -(m+u) \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^{-m-u-1} \mathbf{q}^{n-j} \quad (186)$$

Finally, we return to our main equation,

$$\frac{\partial^{u+1} \mathbf{A}_{-m,n}}{\partial \mathbf{p}} = \frac{\partial}{\partial \mathbf{p}} \left((-1)^u \frac{(m+u-1)!}{(m-1)!} \mathbf{A}_{-m-u,n} \right) = (-1)^{u+1} \frac{(m+u)!}{(m-1)!} \mathbf{A}_{-m-u-1,n} \quad (187)$$

for which works for all $s = u + 1$, and must therefore work for all $s \in \mathbb{Z}_+$. $\square$

A similar theorem is proposed for the multiple partial derivative with respect to the position utilizing the reflection formula.

Theorem 5.15. *For an arbitrarily-quantized operator $\mathbf{A}_{m,-n}$, multiple differentiation with respect to the position gives*

$$\frac{\partial^s \mathbf{A}_{m,-n}}{\partial \mathbf{q}^t} = (-1)^t \frac{(n+t-1)!}{(t-1)!} \mathbf{A}_{m,-n-t} \quad (188)$$

Proof. By mathematical induction, allow our theorem to be true when $t = u$. Then, for $t = u + 1$,

$$\frac{\partial^{u+1} \mathbf{A}_{m,-n}}{\partial \mathbf{q}} = \frac{\partial}{\partial \mathbf{q}} \left((-1)^u \frac{(n+u-1)!}{(n-1)!} \mathbf{A}_{m,-n-t} \right) \quad (189)$$

and by taking the derivative

$$\frac{\partial^{u+1} \mathbf{A}_{m,-n}}{\partial \mathbf{q}} = (-1)^u \frac{(n+u-1)!}{(n-1)!} \lim_{\delta \rightarrow 0} \sum_{j=0}^m \frac{b_j^{(m)}}{\delta} \mathbf{p}^j [(\mathbf{q} + \delta \mathbf{I})^{-n-u} + \mathbf{q}^{-n-u}] \mathbf{p}^{m-j} \quad (190)$$

We turn our attention to the summation. We allow

$$\lim_{\delta \rightarrow 0} \sum_{j=0}^m \frac{b_j^{(m)}}{\delta} \mathbf{p}^j [(\mathbf{q} + \delta \mathbf{I})^{-n-u} + \mathbf{q}^{-n-u}] \mathbf{p}^{m-j} = \lim_{\delta \rightarrow 0} \sum_{j=0}^m \frac{b_j^{(m)}}{\delta} \mathbf{p}^j [\mathbf{q}^{n+u} - (\mathbf{q} + \delta \mathbf{I})^{n+u}] (\mathbf{q}^{-n-u} (\mathbf{q} + \delta \mathbf{I})^{-n-u}) \mathbf{p}^{m-j} \quad (191)$$

and expanding

$$\lim_{\delta \rightarrow 0} \sum_{j=0}^m \frac{b_j^{(m)}}{\delta} \mathbf{p}^j \left[\mathbf{q}^{n+u} - \sum_{k=0}^{n+u} \binom{n+u}{k} \mathbf{q}^{n+u-k} \delta^k \right] (\mathbf{q}^{-n-u} (\mathbf{q} + \delta \mathbf{I})^{-n-u}) \mathbf{p}^{m-j} \quad (192)$$

where we further see that

$$\sum_{j=0}^m b_j^{(m)} \mathbf{p}^j [-(n+u)\mathbf{q}^{n+u-1}] (\mathbf{q}^{-n-u}\mathbf{q}^{-n-u}) \mathbf{p}^{m-j} = -(n+u) \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^{-n-u-1} \mathbf{p}^{m-j} \quad (193)$$

Finally, we return to our main equation,

$$\frac{\partial^{u+1} \mathbf{A}_{m,-n}}{\partial \mathbf{q}} = \frac{\partial}{\partial \mathbf{q}} \left((-1)^u \frac{(n+u-1)!}{(n-1)!} \mathbf{A}_{m,n-u} \right) = (-1)^{u+1} \frac{(n+u)!}{(n-1)!} \mathbf{A}_{m,-n-u-1} \quad (194)$$

for which works for all $t = u + 1$, and must therefore work for all $t \in \mathbb{Z}_+$. $\square$

And finally, we prove for the compatibility of our operators on the invariance of order of differentiation.

Theorem 5.16. *The order of differentiation does not affect the resulting partial derivative. That is, for an arbitrarily-ordered operator $\mathbf{A}_{m,n}$*

$$\frac{\partial^{s+t} \mathbf{A}_{m,n}}{\partial \mathbf{q}^t \partial \mathbf{p}^s} = \frac{\partial^{s+t} \mathbf{A}_{m,n}}{\partial \mathbf{p}^s \partial \mathbf{q}^t} \quad (195)$$

for $s, t \in \mathbb{Z}_+$.

Proof. Recall that for multiple differentiation with respect to the momentum

$$\frac{\partial^s \mathbf{A}_{m,n}}{\partial \mathbf{p}^s} = \frac{m!}{(m-s)!} \mathbf{A}_{m-s,n} \quad (196)$$

and taking the appropriate multiple differentiation with respect to the position

$$\frac{\partial^{s+t} \mathbf{A}_{m,n}}{\partial \mathbf{q}^t \partial \mathbf{p}^s} = \frac{m!}{(m-s)!} \frac{\partial^t \mathbf{A}_{m-s,n}}{\partial \mathbf{q}^t} = \frac{m!}{(m-s)!} \frac{n!}{(n-t)!} \mathbf{A}_{m-s,n-t} \quad (197)$$

Similarly, allow for the multiple differentiation with respect to the position

$$\frac{\partial^t \mathbf{A}_{m,n}}{\partial \mathbf{q}^t} = \frac{n!}{(n-t)!} \mathbf{A}_{m,n-t} \quad (198)$$

and taking the appropriate multiple differentiation with respect to the momentum

$$\frac{\partial^{s+t} \mathbf{A}_{m,n}}{\partial \mathbf{p}^s \partial \mathbf{q}^t} = \frac{n!}{(n-t)!} \frac{\partial^s \mathbf{A}_{m,n-t}}{\partial \mathbf{p}^s} = \frac{n!}{(n-t)!} \frac{m!}{(m-s)!} \mathbf{A}_{m-s,n-t} \quad (199)$$

We see that the differentiation order does affect result.

$$\frac{\partial^{s+t} \mathbf{A}_{m,n}}{\partial \mathbf{q}^t \partial \mathbf{p}^s} = \frac{\partial^{s+t} \mathbf{A}_{m,n}}{\partial \mathbf{p}^s \partial \mathbf{q}^t} \quad (200)$$

The order of $\mathbf{q}$ and $\mathbf{p}$ is irrelevant when dealing with quantum derivatives. This is interesting, as $\mathbf{q}$ and $\mathbf{p}$ do not commute. $\square$

The product rule of differentiation and the rule for the differentiation of a function of a function will no longer be valid, in general. Only when all of the matrices that appear commute with each other will all of the rules of ordinary analysis be valid for this differentiation.

6 Quantization of the Hamilton Equations of Motion

The verification of the properties of the differentiation operation for operators allow us to prove the equivalence of the Hamilton equations of motion in the quantum regime. We provide the commutation relation for an arbitrary quantum image $\mathbf{A}_{m,n}$.

$$[\mathbf{A}_{m,n}, \mathbf{q}] = \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j+1} - \sum_{j=0}^n a_j^{(n)} \mathbf{q}^{j+1} \mathbf{p}^m \mathbf{q}^{n-j} = \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j+1} - \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{q} \mathbf{p}^m \mathbf{q}^{n-j} \quad (201)$$

From the right hand side formulation, we find for the commutator of the momentum with position $\mathbf{q}$, in general,

$$\mathbf{p}^m \mathbf{q} - \mathbf{q} \mathbf{p}^m = -i\hbar m \mathbf{p}^{m-1} \quad (202)$$

from the relation

$$[\mathbf{p}, \mathbf{q}] = \mathbf{p} \mathbf{q} - \mathbf{q} \mathbf{p} = -i\hbar \mathbf{I} \quad (203)$$

By induction, where we say that 203 holds for all $m = k$

$$[\mathbf{p}^k, \mathbf{q}] = \mathbf{p}^k \mathbf{q} - \mathbf{q} \mathbf{p}^k = -ik\hbar \mathbf{p}^{k-1} \quad (204)$$

Now, allow that $m = k + 1$ by

$$\mathbf{p}[\mathbf{p}^k, \mathbf{q}] = \mathbf{p}^{k+1} \mathbf{q} - \mathbf{p} \mathbf{q} \mathbf{p}^k = -ik\hbar \mathbf{p}^k \quad (205)$$

and using 203,

$$\mathbf{p}[\mathbf{p}^k, \mathbf{q}] = \mathbf{p}^{k+1} \mathbf{q} - (\mathbf{q} \mathbf{p} - i\hbar \mathbf{I}) \mathbf{p}^k = -ik\hbar \mathbf{p}^k \quad (206)$$

$$\mathbf{p}^{k+1} \mathbf{q} - \mathbf{q} \mathbf{p}^{k+1} = -i(k+1)\hbar \mathbf{p}^k \quad (207)$$

where the equality holds for all $m = k + 1$ and must therefore hold for all m .

We develop the commutation relation such

$$[\mathbf{A}_{m,n}, \mathbf{q}] = \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j+1} - \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j (\mathbf{p}^m \mathbf{q} + i\hbar m \mathbf{p}^{m-1}) \mathbf{q}^{n-j} \quad (208)$$

$$= \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^m \mathbf{q}^{n-j+1} - \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^m \mathbf{q} \mathbf{q}^{n-j} - i\hbar m \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^{m-1} \mathbf{q}^{n-j} \quad (209)$$

$$[\mathbf{A}_{m,n}, \mathbf{q}] = -i\hbar m \sum_{j=0}^n a_j^{(n)} \mathbf{q}^j \mathbf{p}^{m-1} \mathbf{q}^{n-j} = -i\hbar m \mathbf{A}_{m-1,n} \quad (210)$$

but recall that for a given arbitrary image $\mathbf{A}_{m,n}$,

$$\frac{\partial \mathbf{A}_{m,n}}{\partial \mathbf{p}} = m \mathbf{A}_{m-1,n} \quad (211)$$

and therefore,

$$[\mathbf{A}_{m,n}, \mathbf{q}] = -i\hbar \frac{\partial \mathbf{A}_{m,n}}{\partial \mathbf{p}} \quad (212)$$

Similarly, the commutation relation is read as

$$[\mathbf{A}_{m,n}, \mathbf{p}] = \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j+1} - \sum_{j=0}^m b_j^{(m)} \mathbf{p}^{j+1} \mathbf{q}^n \mathbf{p}^{m-j} \quad (213)$$

and the right hand side can be rewritten using the same procedure as Eq. 203

From the right hand side formulation, we find for the commutator of the position with momentum $\mathbf{p}$, in general,

$$\mathbf{q}^n \mathbf{p} - \mathbf{p} \mathbf{q}^n = i\hbar n \mathbf{q}^{n-1} \quad (214)$$

from the relation

$$[\mathbf{q}, \mathbf{p}] = \mathbf{q} \mathbf{p} - \mathbf{p} \mathbf{q} = i\hbar \mathbf{I} \quad (215)$$

By induction, where we say that 215 holds for all $n = k$

$$[\mathbf{q}^k, \mathbf{p}] = \mathbf{q}^k \mathbf{p} - \mathbf{p} \mathbf{q}^k = ik\hbar \mathbf{q}^{k-1} \quad (216)$$

Now, allow that $n = k + 1$ by

$$\mathbf{q}[\mathbf{q}^k, \mathbf{p}] = \mathbf{q}^{k+1} \mathbf{p} - \mathbf{q} \mathbf{p} \mathbf{q}^k = ik\hbar \mathbf{q}^k \quad (217)$$

and using 215,

$$\mathbf{q}[\mathbf{q}^k, \mathbf{p}] = \mathbf{q}^{k+1} \mathbf{p} - (\mathbf{p} \mathbf{q} - i\hbar \mathbf{I}) \mathbf{q}^k = ik\hbar \mathbf{q}^k \quad (218)$$

$$\mathbf{q}^{k+1} \mathbf{p} - \mathbf{p} \mathbf{q}^{k+1} = i(k+1)\hbar \mathbf{q}^k \quad (219)$$

where the equality holds for all $n = k + 1$ and must therefore hold for all n .

We develop the commutation relation as such

$$[\mathbf{A}_{m,n}, \mathbf{p}] = \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j+1} - \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j (\mathbf{q}^n \mathbf{p} - i\hbar n \mathbf{q}^{n-1}) \mathbf{p}^{m-j} \quad (220)$$

$$= \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^n \mathbf{p}^{m-j+1} - \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^n \mathbf{p} \mathbf{p}^{m-j} + i\hbar n \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^{n-1} \mathbf{p}^{m-j} \quad (221)$$

$$[\mathbf{A}_{m,n}, \mathbf{p}] = i\hbar n \sum_{j=0}^m b_j^{(m)} \mathbf{p}^j \mathbf{q}^{n-1} \mathbf{p}^{m-j} = i\hbar n \mathbf{A}_{m,n-1} \quad (222)$$

but recall that for a given arbitrary image $\mathbf{A}_{m,n}$

$$\frac{\partial \mathbf{A}_{m,n}}{\partial \mathbf{q}} = n \mathbf{A}_{m,n-1} \quad (223)$$

and therefore,

$$[\mathbf{A}_{m,n}, \mathbf{p}] = i\hbar \frac{\partial \mathbf{A}_{m,n}}{\partial \mathbf{q}} \quad (224)$$

We note the following

- Arbitrary ordering (Weyl, Born-Jordan, simplest-symmetrization) satisfies the quantum equations of motion.
- The choice of ordering does not affect the validity of the equations of motion.
- All the possible quantum images of the Hamiltonian will still exhibit the same quantum dynamical behavior and will display the correct quantum image, that is, if the main concern is if it satisfies the equations of motion (which it does).

7 Conclusions

1. The study was able to express the Hamilton equations of motion in the quantum regime.
2. Initially, an equivalence of the Schrodinger and Heisenberg pictures of quantum mechanics was only made possible through Born-Jordan quantization.
3. It was seen, however, that the differentiation procedure (DQFT) laid out by Born and Jordan in their paper eventually leads to terms which are Born-Jordan images.
4. One could easily transform this DFQT by introducing necessary constants and terms (which are normalized) such that under quantization will result in terms which are either Weyl or simplest symmetrization.
5. Whilst the DQST is only applicable to the Born-Jordan quantization, through the Theorems, it was shown that any arbitrary quantization will produce a derivative in that arbitrary quantization's image.
6. The construction of the Hamilton equations of motion using any of the quantization rules will result to the correct quantum image. The dynamics of the operators follow the same behavior as the classical counterpart.

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APPENDIX

We introduce the problem of the harmonic oscillator utilizing the quantization images shown here today. In the classical case, the harmonic oscillator is written as

$$H = \frac{1}{2\mu}p^2 + \frac{1}{2}\mu\omega^2q^2 \quad (225)$$

for mass μ and frequency ω .

Weyl quantization. Using Weyl quantization, the variables are transformed such that

$$p \mapsto \mathbf{T}_{1,0} \quad q \mapsto \mathbf{T}_{0,1} \quad (226)$$

and 225 is rewritten as

$$\mathbf{H}_W = \frac{1}{2\mu}\mathbf{T}_{2,0} + \frac{1}{2}\mu\omega^2\mathbf{T}_{0,2} \quad (227)$$

and finding the commutator of the Hamiltonian with position gives

$$[\mathbf{H}_W, \mathbf{T}_{0,1}] = \left[\frac{1}{2\mu}\mathbf{T}_{2,0}\mathbf{T}_{0,1} + \frac{1}{2}\mu\omega^2\mathbf{T}_{0,2}\mathbf{T}_{0,1} \right] - \left[\frac{1}{2\mu}\mathbf{T}_{0,1}\mathbf{T}_{2,0} + \frac{1}{2}\mu\omega^2\mathbf{T}_{0,1}\mathbf{T}_{0,2} \right] = \frac{1}{2\mu}[\mathbf{T}_{2,0}, \mathbf{T}_{0,1}] - \frac{1}{2}\mu\omega^2[\mathbf{T}_{0,2}, \mathbf{T}_{0,1}] \quad (228)$$

[2] writes the explicit commutation relation for all quantization types. For Weyl quantization,

$$[\mathbf{T}_{m,n}, \mathbf{T}_{r,s}] = 2 \sum_{j=0}^{\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} (-1)^k \binom{2j+1}{k} \frac{m!}{(m-k)!} \frac{n!}{(n-2j-1+k)!} \times \frac{r!}{(r-2j-1+k)!} \frac{s!}{(s-k)!} \mathbf{T}_{m+r-2j-1, n+s-2j-1} \quad (229)$$

where, for $m = 2$, $s = 1$, $n, r = 0$, we find

$$[\mathbf{T}_{2,0}, \mathbf{T}_{0,1}] = 2 \sum_{j=0}^{\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} (-1)^k \binom{2j+1}{k} \frac{2!}{(2-k)!} \frac{1}{(-2j-1+k)!} \frac{1}{(-2j-1+k)!} \frac{1!}{(1-k)!} \mathbf{T}_{2-2j, -2j-1} \quad (230)$$

In this case, the value of k must not be so that it would produce negative factorials. We set $k = 1$ as the maximum, letting $2j+1 = 0$ and $j = 0$.

$$[\mathbf{T}_{2,0}, \mathbf{T}_{0,1}] = 2(i\hbar/2)(-1) \frac{2!}{(1)!} \frac{1}{(0)!} \frac{1}{(0)!} \frac{1!}{(0)!} \mathbf{T}_{1,0} = -i2\hbar\mathbf{T}_{1,0} \quad (231)$$

When $m, s = 0$, $n = 1$, and $r = 2$,

$$[\mathbf{T}_{0,1}, \mathbf{T}_{2,0}] = 2 \sum_{j=0}^{\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} (-1)^k \binom{2j+1}{k} \frac{0!}{(-k)!} \frac{1!}{(-2j+k)!} \frac{2!}{(1-2j+k)!} \frac{0!}{(-k)!} \mathbf{T}_{1-2j, -2j} \quad (232)$$

for which k could only be zero otherwise negative factorials would appear. As j could only be defined for integers, this commutator would need to be set to 0. Finally,

$$[\mathbf{H}_W, \mathbf{T}_{0,1}] = \frac{1}{2\mu} [\mathbf{T}_{2,0}, \mathbf{T}_{0,1}] - \frac{1}{2} \mu \omega^2 [\mathbf{T}_{0,1}, \mathbf{T}_{2,0}] = -\frac{i\hbar}{\mu} \mathbf{T}_{1,0} \quad (233)$$

By the Theorems, the derivative here is given

$$\frac{\partial \mathbf{H}_W}{\partial \mathbf{p}} = \frac{1}{\mu} \mathbf{T}_{1,0} \quad (234)$$

and finally an equality is found

$$[\mathbf{H}_W, \mathbf{q}] = -i\hbar \frac{\partial \mathbf{H}_W}{\partial \mathbf{p}} \quad (235)$$

Similarly, we find the commutation relation with the momentum operator.

$$[\mathbf{H}_W, \mathbf{T}_{1,0}] = \frac{1}{2\mu} [\mathbf{T}_{2,0}, \mathbf{T}_{1,0}] + \frac{1}{2} \mu \omega^2 [\mathbf{T}_{0,2}, \mathbf{T}_{1,0}] \quad (236)$$

and the commutation relations are written

$$[\mathbf{T}_{2,0}, \mathbf{T}_{1,0}] = 2 \sum_{j=0}^{\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} (-1)^k \binom{2j+1}{k} \frac{2!}{(2-k)!} \frac{1}{(-2j-1+k)!} \frac{1}{(-2j+k)!} \frac{1}{(-k)!} \mathbf{T}_{2-2j, -2j} \quad (237)$$

by which would only result in real values if $k = 0$. As the value of j is an integer, j forces the commutator to be 0.

$$[\mathbf{T}_{2,0}, \mathbf{T}_{1,0}] = 0 \quad (238)$$

Further,

$$[\mathbf{T}_{0,2}, \mathbf{T}_{1,0}] = 2 \sum_{j=0}^{\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} (-1)^k \binom{2j+1}{k} \frac{1}{(-k)!} \frac{2!}{(1-2j+k)!} \frac{1}{(-2j+k)!} \frac{1}{(-k)!} \mathbf{T}_{-2j, 1-2j} \quad (239)$$

by which we are forced to say that $k = 0$ for the commutator to be real valued.

$$[\mathbf{T}_{0,2}, \mathbf{T}_{1,0}] = 2 \sum_{j=0}^{\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} (2j+1) \frac{2!}{(1-2j)!} \frac{1}{(-2j)!} \mathbf{T}_{-2j, 1-2j} \quad (240)$$

and further set $j = 0$ such that the factorials are well-behaved.

$$[\mathbf{T}_{0,2}, \mathbf{T}_{1,0}] = 2 \frac{i\hbar}{2} \frac{2!}{1!} \frac{1}{0!} \mathbf{T}_{0,1} = 2i\hbar \mathbf{T}_{0,1} \quad (241)$$

We then write for Eq. 236

$$[\mathbf{H}_W, \mathbf{T}_{1,0}] = i\hbar \mu \omega^2 \mathbf{T}_{0,1} \quad (242)$$

In a similar fashion we compute for the derivative of $\mathbf{H}_W$ with respect to the position

$$\frac{\partial \mathbf{H}_W}{\partial \mathbf{q}} = \mu \omega^2 \mathbf{T}_{0,1} \quad (243)$$

and find the equality to be

$$[\mathbf{H}_W, \mathbf{p}] = i\hbar \frac{\partial \mathbf{H}_W}{\partial \mathbf{q}} \quad (244)$$

Simplest symmetrization. The variables are transformed such that

$$p \mapsto \mathbf{S}_{1,0} \quad q \mapsto \mathbf{S}_{0,1} \quad (245)$$

and that 225 is rewritten as

$$\mathbf{H}_S = \frac{1}{2\mu}\mathbf{S}_{2,0} + \frac{1}{2}\mu\omega^2\mathbf{S}_{0,2} \quad (246)$$

and the commutation relation to be

$$[\mathbf{H}_S, \mathbf{S}_{0,1}] = \frac{1}{2\mu}[\mathbf{S}_{2,0}, \mathbf{S}_{0,1}] + \frac{1}{2}\mu\omega^2[\mathbf{S}_{0,2}, \mathbf{S}_{0,1}] \quad (247)$$

The commutation relation is written by [2] for simplest symmetrization

$$\begin{aligned} [\mathbf{S}_{m,n}, \mathbf{S}_{r,s}] &= 2 \sum_{j=0}^{+\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} \binom{2j+1}{k} (-1)^k \sum_{\ell=0}^{+\infty} \frac{(i\hbar/2)^{2\ell}}{(2\ell)!} \sum_{t=0}^{+\infty} \frac{(i\hbar/2)^{2t}}{(2t)!} \frac{m!}{(m-k-2\ell)!} \frac{n!}{(n-2j-1+k-2\ell)!} \\ &\times \frac{r!}{(r-2j-1+k-2t)!} \frac{s!}{(s-k-2t)!} \sum_{u=0}^{+\infty} \frac{(-i\hbar/2)^u E_u}{u!} \frac{(m+r-2j-1-2\ell-2t)!}{(m+r-2j-1-2\ell-2t-u)!} \\ &\times \frac{(n+s-2j-1-2\ell-2t)!}{(n+s-2j-1-2\ell-2t-u)!} \mathbf{S}_{m+r-2j-1-2\ell-2t-u, n+s-2j-1-2\ell-2t-u} \end{aligned} \quad (248)$$

and thus

$$\begin{aligned} [\mathbf{S}_{2,0}, \mathbf{S}_{0,1}] &= 2 \sum_{j=0}^{+\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} \binom{2j+1}{k} (-1)^k \sum_{\ell=0}^{+\infty} \frac{(i\hbar/2)^{2\ell}}{(2\ell)!} \sum_{t=0}^{+\infty} \frac{(i\hbar/2)^{2t}}{(2t)!} \frac{2!}{(2-k-2\ell)!} \frac{1}{(-2j-1+k-2\ell)!} \\ &\times \frac{1}{(-2j-1+k-2t)!} \frac{1}{(1-k-2t)!} \sum_{u=0}^{+\infty} \frac{(-i\hbar/2)^u E_u}{u!} \frac{(1-2j-2\ell-2t)!}{(1-2j-2\ell-2t-u)!} \\ &\times \frac{(-2j-2\ell-2t)!}{(-2j-2\ell-2t-u)!} \mathbf{S}_{1-2j-2\ell-2t-u, -2j-2\ell-2t-u} \end{aligned} \quad (249)$$

where for values of $j, \ell, t, u > 0$ will result in negative factorials. We set $j, \ell, t, u = 0$. In the case for k , we set $k = 1$.

$$[\mathbf{S}_{2,0}, \mathbf{S}_{0,1}] = -2i\hbar\mathbf{S}_{1,0} \quad (250)$$

Similarly, the commutation relation is written

$$\begin{aligned} [\mathbf{S}_{0,2}, \mathbf{S}_{0,1}] &= 2 \sum_{j=0}^{+\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} \binom{2j+1}{k} (-1)^k \sum_{\ell=0}^{+\infty} \frac{(i\hbar/2)^{2\ell}}{(2\ell)!} \sum_{t=0}^{+\infty} \frac{(i\hbar/2)^{2t}}{(2t)!} \frac{1}{(-k-2\ell)!} \frac{2}{(1-2j+k-2\ell)!} \\ &\times \frac{1}{(-2j-1+k-2t)!} \frac{1}{(1-k-2t)!} \sum_{u=0}^{+\infty} \frac{(-i\hbar/2)^u E_u}{u!} \frac{(-2j-1-2\ell-2t)!}{(-2j-1-2\ell-2t-u)!} \\ &\times \frac{(2-2j-2\ell-2t)!}{(2-2j-2\ell-2t-u)!} \mathbf{S}_{-2j-1-2\ell-2t-u, 2-2j-2\ell-2t-u} = 0 \end{aligned} \quad (251)$$

where from the term $\frac{(-2j-1-2\ell-2t)!}{(-2j-1-2\ell-2t-u)!}$, for the factorials to be well-behaved, we set the whole commutator to 0.

The same pattern of setting to 0 will occur if for either pair m, r and n, s are both zero to avoid retaining a $(-1)!$ term in the latter fractions. This allows us to declare $[\mathbf{S}_{2,0}, \mathbf{S}_{1,0}] = 0$

For completeness, let's find $[\mathbf{S}_{0,2}, \mathbf{S}_{1,0}]$:

$$\begin{aligned} [\mathbf{S}_{0,2}, \mathbf{S}_{1,0}] &= 2 \sum_{j=0}^{+\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} \binom{2j+1}{k} (-1)^k \sum_{\ell=0}^{+\infty} \frac{(i\hbar/2)^{2\ell}}{(2\ell)!} \sum_{t=0}^{+\infty} \frac{(i\hbar/2)^{2t}}{(2t)!} \frac{1}{(-k-2\ell)!} \frac{2}{(1-2j+k-2\ell)!} \\ &\times \frac{1}{(-2j+k-2t)!} \frac{1}{(-k-2t)!} \sum_{u=0}^{+\infty} \frac{(-i\hbar/2)^u E_u}{u!} \frac{(-2j-2\ell-2t)!}{(-2j-2\ell-2t-u)!} \\ &\times \frac{(1-2j-2\ell-2t)!}{(1-2j-2\ell-2t-u)!} \mathbf{S}_{-2j-2\ell-2t-u, 1-2j-2\ell-2t-u} \end{aligned} \quad (252)$$

where to avoid negative factorials, we set $j, k, \ell, t, u = 0$.

$$[\mathbf{S}_{0,2}, \mathbf{S}_{1,0}] = 2i\hbar \mathbf{S}_{0,1} \quad (253)$$

and so it is written

$$[\mathbf{H}_S, \mathbf{S}_{0,1}] = -\frac{1}{\mu} i\hbar \mathbf{S}_{1,0} \quad (254)$$

and finding the partial derivative with respect to the momentum

$$\frac{\partial \mathbf{H}_S}{\partial \mathbf{p}} = \frac{1}{\mu} \mathbf{S}_{1,0} \quad (255)$$

and that

$$[\mathbf{H}_S, \mathbf{q}] = -i\hbar \frac{\partial \mathbf{H}_S}{\partial \mathbf{p}} \quad (256)$$

In a similar manner, by finding the commutation relation with the momentum

$$[\mathbf{H}_S, \mathbf{S}_{1,0}] = \frac{1}{2\mu} [\mathbf{S}_{2,0}, \mathbf{S}_{1,0}] + \frac{1}{2} \mu \omega^2 [\mathbf{S}_{0,2}, \mathbf{S}_{1,0}] \quad (257)$$

where the commutation relations are previously found. Thus

$$[\mathbf{H}_S, \mathbf{S}_{1,0}] = \mu \omega^2 i\hbar \mathbf{S}_{0,1} \quad (258)$$

and finding the derivative with respect to the position

$$\frac{\partial \mathbf{H}_S}{\partial \mathbf{q}} = \mu \omega^2 \mathbf{S}_{0,1} \quad (259)$$

Finally,

$$[\mathbf{H}_S, \mathbf{p}] = i\hbar \frac{\partial \mathbf{H}_S}{\partial \mathbf{q}} \quad (260)$$

Born-Jordan quantization. The variables are transformed such that

$$p \mapsto \mathbf{B}_{1,0} \quad q \mapsto \mathbf{B}_{0,1} \quad (261)$$

and thus

$$[\mathbf{H}_{BJ}, \mathbf{B}_{0,1}] = \frac{1}{2\mu} [\mathbf{B}_{2,0}, \mathbf{B}_{0,1}] + \frac{1}{2} \mu \omega^2 [\mathbf{B}_{0,2}, \mathbf{B}_{0,1}] \quad (262)$$

for which the commutation relations are written by [2]

$$\begin{aligned} [\mathbf{B}_{m,n}, \mathbf{B}_{r,s}] &= 2 \sum_{j=0}^{+\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} \binom{2j+1}{k} (-1)^k \sum_{\ell=0}^{+\infty} \frac{(i\hbar/2)^{2\ell}}{(2\ell+1)!} \sum_{t=0}^{+\infty} \frac{(i\hbar/2)^{2t}}{(2t+1)!} \frac{m!}{(m-k-2\ell)!} \frac{n!}{(n-2j-1+k-2\ell)!} \\ &\times \frac{r!}{(r-2j-1+k-2t)!} \frac{s!}{(s-k-2t)!} \sum_{u=0}^{+\infty} \frac{(-i\hbar)^u B_u(1/2)}{u!} \frac{(m+r-2j-1-2\ell-2t)!}{(m+r-2j-1-2\ell-2t-u)!} \\ &\times \frac{(n+s-2j-1-2\ell-2t)!}{(n+s-2j-1-2\ell-2t-u)!} \mathbf{B}_{m+r-2j-1-2\ell-2t-u, n+s-2j-1-2\ell-2t-u} \end{aligned} \quad (263)$$

which is in the same form of the commutation relation for the simplest symmetrization, and thus possessing the same weakness when either pair m, r and n, s are set both to 0. We find

$$\begin{aligned} [\mathbf{B}_{2,0}, \mathbf{B}_{0,1}] &= 2 \sum_{j=0}^{+\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} \binom{2j+1}{k} (-1)^k \sum_{\ell=0}^{+\infty} \frac{(i\hbar/2)^{2\ell}}{(2\ell+1)!} \sum_{t=0}^{+\infty} \frac{(i\hbar/2)^{2t}}{(2t+1)!} \frac{2}{(2-k-2\ell)!} \frac{1}{(-2j-1+k-2\ell)!} \\ &\times \frac{1}{(-2j-1+k-2t)!} \frac{1}{(1-k-2t)!} \sum_{u=0}^{+\infty} \frac{(-i\hbar)^u B_u(1/2)}{u!} \frac{(1-2j-2\ell-2t)!}{(1-2j-2\ell-2t-u)!} \\ &\times \frac{(-2j-2\ell-2t)!}{(-2j-2\ell-2t-u)!} \mathbf{B}_{1-2j-2\ell-2t-u, -2j-2\ell-2t-u} \end{aligned} \quad (264)$$

and we set $j, \ell, t, u = 0$ and $k = 1$

$$[\mathbf{B}_{2,0}, \mathbf{B}_{0,1}] = -2i\hbar \mathbf{B}_{1,0} \quad (265)$$

and for $[\mathbf{B}_{0,2}, \mathbf{B}_{0,1}]$

$$\begin{aligned} [\mathbf{B}_{0,2}, \mathbf{B}_{0,1}] &= 2 \sum_{j=0}^{+\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} \binom{2j+1}{k} (-1)^k \sum_{\ell=0}^{+\infty} \frac{(i\hbar/2)^{2\ell}}{(2\ell+1)!} \sum_{t=0}^{+\infty} \frac{(i\hbar/2)^{2t}}{(2t+1)!} \frac{1}{(-k-2\ell)!} \frac{2}{(1-2j+k-2\ell)!} \\ &\quad \times \frac{1}{(-2j-1+k-2t)!} \frac{1}{(1-k-2t)!} \sum_{u=0}^{+\infty} \frac{(-i\hbar)^u B_u(1/2)}{u!} \frac{(-2j-1-2\ell-2t)!}{(-2j-1-2\ell-2t-u)!} \\ &\quad \times \frac{(2-2j-2\ell-2t)!}{(2-2j-2\ell-2t-u)!} \mathbf{B}_{-2j-1-2\ell-2t-u, 2-2j-2\ell-2t-u} \end{aligned} \quad (266)$$

where the latter fractions would behave badly due to the $(-1)!$. Thus we set this and the commutator $[\mathbf{B}_{2,0}, \mathbf{B}_{1,0}] = 0$. For completeness, let's find $[\mathbf{B}_{0,2}, \mathbf{B}_{1,0}]$

$$\begin{aligned} [\mathbf{B}_{0,2}, \mathbf{B}_{1,0}] &= 2 \sum_{j=0}^{+\infty} \frac{(i\hbar/2)^{2j+1}}{(2j+1)!} \sum_{k=0}^{2j+1} \binom{2j+1}{k} (-1)^k \sum_{\ell=0}^{+\infty} \frac{(i\hbar/2)^{2\ell}}{(2\ell+1)!} \sum_{t=0}^{+\infty} \frac{(i\hbar/2)^{2t}}{(2t+1)!} \frac{1}{(-k-2\ell)!} \frac{2}{(-2j+k-2\ell)!} \\ &\quad \times \frac{1}{(-2j+k-2t)!} \frac{1}{(-k-2t)!} \sum_{u=0}^{+\infty} \frac{(-i\hbar)^u B_u(1/2)}{u!} \frac{(-2j-2\ell-2t)!}{(-2j-2\ell-2t-u)!} \\ &\quad \times \frac{(1-2j-2\ell-2t)!}{(1-2j-2\ell-2t-u)!} \mathbf{B}_{-2j-2\ell-2t-u, 1-2j-2\ell-2t-u} \end{aligned} \quad (267)$$

where only by setting all $j, k, \ell, t, u = 0$ will we find

$$[\mathbf{B}_{0,2}, \mathbf{B}_{1,0}] = 2i\hbar \mathbf{B}_{0,1} \quad (268)$$

and thus we write, for the commutation relation for $\mathbf{H}_{BJ}$ for both position and momentum

$$[\mathbf{H}_{BJ}, \mathbf{B}_{0,1}] = -\frac{1}{\mu} i\hbar \mathbf{B}_{1,0} \quad [\mathbf{H}_{BJ}, \mathbf{B}_{1,0}] = \mu\omega^2 i\hbar \mathbf{B}_{0,1} \quad (269)$$

and derivatives with respect to momentum and position are

$$\frac{\partial \mathbf{H}_{BJ}}{\partial \mathbf{p}} = \frac{1}{\mu} \mathbf{B}_{1,0} \quad \frac{\partial \mathbf{H}_{BJ}}{\partial \mathbf{q}} = \mu\omega^2 \mathbf{B}_{0,1} \quad (270)$$

and finally

$$[\mathbf{H}_{BJ}, \mathbf{B}_{0,1}] = -i\hbar \frac{\partial \mathbf{H}_{BJ}}{\partial \mathbf{p}} \quad [\mathbf{H}_{BJ}, \mathbf{B}_{1,0}] = i\hbar \frac{\partial \mathbf{H}_{BJ}}{\partial \mathbf{q}} \quad (271)$$

From the results, we see that all arbitrary quantizations satisfy the Hamilton equations of motion.